RV College of Engineering

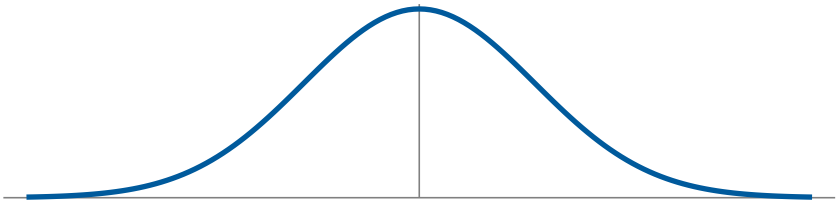
Department of Industrial Engineering and Management

Statistics for Data Analytics

Course Code: IM241AT

Semester IV • Professional Core Course

Unit 5



A Student Compilation for Study and Reference

Semester: IV						
STATISTICS FOR DATA ANALYTICS **Category: Professional Core Course** **(Theory)**						
Course Code	**:**	**IM241AT**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**3.00 Hours**

Unit-I	**09 Hrs**
Data Summary and Presentation: Data types, tabular and graphical displays: Stem and Leaf diagrams, Histograms, Box plots, Radar diagrams. **Concepts of Probability:** Sample spaces and Events, Interpretations of probability, Addition rules, Conditional probability, Multiplication and Total probability rules, Independence, Bayes Theorem. Numerical Problems	
Unit-II	**09 Hrs**
Concepts of Random Variables: Random Variables, Discrete and continuous random variables. Probability distributions and mass functions, Numerical Problems **Discrete Probability Distributions:** Discrete uniform, Binominal, Poisson, Geometric, Negative binomial, Applications, Numerical Problems.	
Unit-III	**09 Hrs**
Continuous Probability Distributions: Continuous uniform, Normal, Normal approximations, Exponential, Applications, Numerical Problems. **Sampling Distribution:** Sampling distribution, Central Limit Theorem, Sampling distribution of means and derived quantities, Numerical Problems.	
Unit-IV	**09 Hrs**
Simple Linear Regression and Correlation: Empirical models, Simple Linear Regression, Properties of Least square Estimators and Estimation of variances, Common abuses of regression, Prediction of new observations, Correlation, Numerical Problems. Interpretation of graphical output from software packages such as Minitab	
Unit-V	**09 Hrs**
Statistical Inference for a single sample: Hypothesis testing, Inference on the mean of a normal population (variance known and unknown), Inference on the variance of a Normal population, Testing for Goodness of Fit, Tests of association, Numerical Problems **Interval Estimation:** Confidence Intervals on mean (variance known and unknown), and variance of a normal population	

Course Outcomes: After completing the course, the students will be able to	
CO1	Describe and report data set using data analysis, presentation and interpretation techniques to understand various phenomena in the fields of science and engineering.
CO2	Apply various statistical processing techniques to handle a set of data to estimate probabilities.
CO3	Apply an appropriate statistical tool and analyze a specific set of data to estimate and draw conclusions about population parameters
CO4	Draw inferences about population parameters and relations between variables based on analysis of sample data

References Books	
1.	Engineering Statistics, Douglas C. Montgomery, George C. Runger, Norma FarisHubele, 5th Edition, 2011, John Wiley & Sons, Inc., ISBN-13: 978- 0-470-63147-8
2.	Applied statistics and Probability for Engineers, Douglas C Montgomery, George C Runger, Wiley, 4th Edition, 2007, Asia Student Edition, ISBN: 978-81-265-2315-3.
3.	Statistics for Management, Richard I Levin, David S Rubin, 7th Edition, 1997, Prentice Hall India, ISBN: 9780134762920.
4.	Probability and Statistics for Engineers and Scientists, Walpole, Myers, Myers, Ye, 8th Edition, 2007, Pearson Education Inc., ISBN: 978-81-317-1552-9.
5.	Softwares : Microsoft Excel / Minitab / Matlab / R

6.	Online resources: a) http://172.16.44.44/nnptel.html - choose NOC:Introduction to Data Analytics(Course sponsored by Aricent) b) https://onlinecourses.nptel.ac.in/noc22_mg31/preview -Introduction to probability and Statistics c) https://newonlinecourses.science.psu.edu/statprogram/undergraduate-studies d) https://www.khanacademy.org/math/statistics-probability

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) **ADDING UPTO 40**	**40**
MAXIMUM MARKS FOR THE CIE THEORY		**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1: (Compulsory)	16
3 & 4	Unit 2: (Internal Choice)	16
5 & 6	Unit 3: (Internal Choice)	16
7 & 8	Unit 4: (Internal Choice)	16
9 & 10	Unit 5: (Internal Choice)	16
	TOTAL	**100**

A Note on This Compilation

This book is **not** the official RV College of Engineering notes. It is a compilation of student notes together with the handwritten notes provided by **Dr. Ramaa A Nadig**, Department of Industrial Engineering and Management.

All information contained in this book has been drawn from her notes and lectures. The material has been organised, typeset, and arranged to follow the prescribed course syllabus, but it should be treated as a supplementary study aid rather than an official course document.

Every effort has been made to reproduce the source notes faithfully and to preserve their content without addition or deletion. Where the original handwritten working was unclear, the final answers recorded in the notes were treated as authoritative. Readers should always cross-check against the prescribed textbooks and their instructor's guidance.

Compiled from the notes and lectures of Dr. Ramaa A Nadig, Department of Industrial Engineering and Management.

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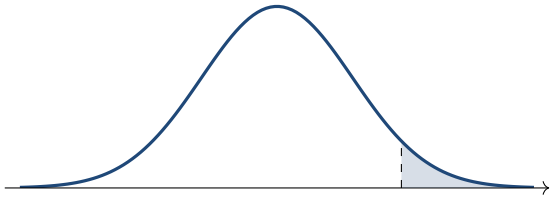
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Unit 5



Chapter 1

Introduction to Hypothesis Testing

Many engineering problems require that we decide whether to accept or reject a statement about some parameter. The statement is called a *hypothesis*, and the decision-making procedure about the hypothesis is called *hypothesis testing*. This chapter builds the conceptual machinery — null and alternative hypotheses, the two kinds of error, the significance level, power, and P-values — that every later test in this unit relies upon.

1.1 Statistical Hypotheses

Statistical Hypothesis

A **statistical hypothesis** is a statement about the parameters of one or more populations.

Since we use probability distributions to represent populations, a statistical hypothesis may also be thought of as a statement about the probability distribution of a random variable. The hypothesis will usually involve one or more parameters of this distribution.

For example, suppose we are interested in the burning rate of a solid propellant used to power aircrew escape systems. The burning rate is a random variable that can be described by a probability distribution. Suppose our interest focuses on the *mean* burning rate (a parameter of this distribution). Specifically, we are interested in deciding whether or not the mean burning rate is $50\,\mathrm{cm\,s^{-1}}$. We may express this as

$$H_0:\ \mu = 50\,\mathrm{cm\,s^{-1}}, \qquad H_1:\ \mu \neq 50\,\mathrm{cm\,s^{-1}}.$$

Null and Alternative Hypotheses

The statement $H_0 : \mu = 50\,\mathrm{cm\,s^{-1}}$ is called the **null hypothesis**, and the statement $H_1 : \mu \neq 50\,\mathrm{cm\,s^{-1}}$ is called the **alternative hypothesis**.

Because the alternative hypothesis specifies values of μ that could be either greater than or less than $50\,\mathrm{cm\,s^{-1}}$, it is called a *two-sided alternative hypothesis*. In some situations we may wish to formulate a *one-sided alternative hypothesis*, as in

$$H_0:\ \mu = 50\,\mathrm{cm\,s^{-1}} \qquad H_0:\ \mu = 50\,\mathrm{cm\,s^{-1}}$$
$$H_1:\ \mu < 50\,\mathrm{cm\,s^{-1}} \quad \text{or} \quad H_1:\ \mu > 50\,\mathrm{cm\,s^{-1}}.$$

Important

Hypotheses are always statements about the population or distribution under study, *not* statements about the sample.

How the null value is chosen

The value of the population parameter specified in the null hypothesis (here $50\,\mathrm{cm\,s^{-1}}$) is usually determined in one of three ways.

1. It may result from past experience or knowledge of the process, or from previous tests or experiments. The objective of hypothesis testing is then usually to determine whether the parameter value has changed.
2. It may be determined from some theory or model regarding the process under study. Here the objective of hypothesis testing is to verify the theory or model.
3. It may result from external considerations, such as design or engineering specifications, or from contractual obligations. In this case the objective of hypothesis testing is *conformance testing*.

A procedure leading to a decision about a particular hypothesis is called a *test of a hypothesis*. Hypothesis-testing procedures rely on using the information in a random sample from the population of interest. If this information is consistent with the hypothesis, we will not reject the hypothesis; however, if this information is inconsistent with the hypothesis, we will conclude that the hypothesis is false.

1.2 The Test Statistic and the Critical Region

In the propellant example, suppose that a sample of $n = 10$ specimens is tested and the sample mean burning rate $\bar{x}$ is observed. The sample mean is an estimate of the true population mean μ. A value of the sample mean $\bar{x}$ that falls close to the hypothesized value $\mu = 50\,\mathrm{cm\,s^{-1}}$ does not conflict with the null hypothesis that the true mean μ is really $50\,\mathrm{cm\,s^{-1}}$. On the other hand, a sample mean that is considerably different from $50\,\mathrm{cm\,s^{-1}}$ is evidence in support of the alternative hypothesis H_1. Thus the sample mean is the *test statistic* in this case.

The sample mean can take on many different values. Suppose that

- if $48.5 \leq \bar{x} \leq 51.5$, we will *not* reject the null hypothesis $H_0 : \mu = 50$; and
- if either $\bar{x} < 48.5$ or $\bar{x} > 51.5$, we will reject the null hypothesis in favour of the alternative hypothesis $H_1 : \mu \neq 50$,

as illustrated in Figure 1.1.

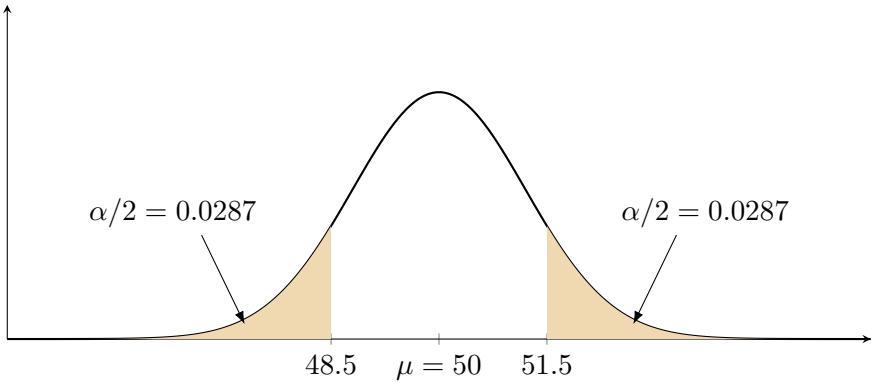


Figure 1.1: Critical region for $H_0 : \mu = 50$ versus $H_1 : \mu \neq 50$ with $n = 10$.

Critical Region, Acceptance Region and Critical Values

The values of $\bar{x}$ that are < 48.5 or > 51.5 constitute the **critical region** (or rejection region) for the test. All values in the interval $48.5 \leq \bar{x} \leq 51.5$ form a region for which we fail to reject the null hypothesis, called the **acceptance region**. The boundaries between the critical and acceptance regions are called the **critical values** (here 48.5 and 51.5).

It is customary to state conclusions relative to the null hypothesis H_0. Therefore, we reject H_0 in favour of H_1 if the test statistic falls in the critical region, and fail to reject H_0 otherwise.

1.3 Type I and Type II Errors

This decision procedure can lead to either of two wrong conclusions.

Type I and Type II Errors

Type I error: rejecting the null hypothesis H_0 when it is true.

Type II error: failing to reject the null hypothesis H_0 when it is false.

The four possible outcomes are summarised in Table 1.1.

Table 1.1: Decisions in hypothesis testing.

Decision	**H_0 is True**	**H_0 is False**
Fail to reject H_0	No error	Type II error
Reject H_0	Type I error	No error

Significance Level α

The probability of making a Type I error is denoted by α:

$$\alpha = \mathrm{P}(\text{Type I error}) = \mathrm{P}(\text{reject } H_0 \mid H_0 \text{ true}).$$

Sometimes the Type I error probability is called the *significance level*, the *α-error*, or the *size of the test*.

Note

1. We can reduce α by widening the acceptance region.
2. We can also reduce α by increasing the sample size.

Type II Error Probability β

The probability of a Type II error is denoted by β:

$$\beta = \text{P(Type II error)} = \text{P(fail to reject } H_0 \mid H_0 \text{ false)}.$$

To calculate β we must have a specific alternative hypothesis, i.e. a particular value of μ for H_1.

Note

1. The probability of making a Type II error, β, increases rapidly as the true value of μ approaches the hypothesized value.
2. The Type II error probability also depends on the sample size n. Increasing the sample size results in a decrease in the probability of a Type II error.

1.4 Four Important Points to Ponder

1. The size of the critical region, and consequently the probability of a Type I error α, can always be reduced by appropriate selection of the critical values.
2. Type I and Type II errors are related. A decrease in the probability of one type of error always results in an increase in the probability of the other, provided that the sample size n does not change.
3. An increase in sample size reduces β, provided that α is held constant.
4. When the null hypothesis is false, β increases as the true value of the parameter approaches the value hypothesized in the null hypothesis. The value of β decreases as the difference between the true mean and the hypothesized value increases.

1.5 Strong versus Weak Conclusions

Generally the analyst controls the Type I error probability α when he or she selects the critical values. Thus it is easy for the analyst to set the Type I error probability at any desired value. Since the analyst can directly control the probability of wrongly rejecting H_0, we always think of rejection of the null hypothesis H_0 as a *strong conclusion*.

On the other hand, the probability of a Type II error β is not constant but depends on the true value of the parameter. It also depends on the sample size we have selected. Since β is a function of both the sample size and the extent to which the null hypothesis H_0 is false, it is customary to think of the decision to accept H_0 as a *weak conclusion*, unless we know that β is acceptably small. Therefore, rather than saying "accept H_0", we prefer the terminology "fail to reject H_0". Failing to reject H_0 implies that we have not found sufficient evidence to reject H_0, i.e. to make a strong statement. Failing to reject H_0 does not necessarily mean that there is a high probability that H_0 is true; it may simply mean that more data are required to reach a strong conclusion.

1.6 Power of a Test

Power

The **power** of a statistical test is the probability of rejecting the null hypothesis H_0 when the alternative hypothesis is true.

The power is computed as $1-\beta$, and can be interpreted as the probability of correctly rejecting a false null hypothesis. We often compare statistical tests by comparing their power properties. For example, if we are considering two tests and one has a higher power than the other, we would prefer the test with the higher power (assuming other properties are comparable).

1.7 One-Sided and Two-Sided Hypotheses

In formulating one-sided alternative hypotheses, remember that rejecting H_0 is always a strong conclusion. Consequently, we should put the statement about which it is important to make a strong conclusion in the alternative hypothesis. In real-world problems, this will often depend on our point of view and experience with the situation.

1.8 P-values in Hypothesis Tests

P-value

The **P-value** is the smallest level of significance that would lead to rejection of the null hypothesis H_0 with the given data.

It is not always easy to compute the exact P-value for a test. However, most modern computer programs for statistical analysis report P-values.

1.9 General Procedure for Hypothesis Tests

Eight-Step Procedure

1. From the problem context, identify the parameter of interest.
2. State the null hypothesis H_0.
3. Specify an appropriate alternative hypothesis H_1.
4. Choose a significance level α.
5. Determine an appropriate test statistic.
6. State the rejection region for the statistic. (This step is not necessary if the P-value approach is used.)
7. Compute any necessary sample quantities, substitute these into the equation for the test statistic, and compute that value.
8. Decide whether or not H_0 should be rejected and report that in the problem context.

1.10 Worked Problems

Example 1.1 (Correctly stated hypotheses)**.** In each of the following situations, state whether it is a correctly stated hypothesis-testing problem and why.

(a) $H_0 : \mu = 25,\ H_1 : \mu \neq 25.$
Yes. We will always state H_0 as an equality involving the specific value of the parameter.

(b) $H_0 : \sigma > 10,\ H_1 : \sigma = 10.$
No. The null hypothesis should specify a value (equality); the alternative should not.

(c) $H_0 : \bar{x} = 50,\ H_1 : \bar{x} \neq 50.$
No. Hypothesis statements are for the population, not for the sample statistic $\bar{x}$.

(d) $H_0 : p = 0.1,\ H_1 : p = 0.5.$
No. The values in the two hypotheses are different; the null and alternative must refer to the same parameter value.

(e) $H_0 : s = 30,\ H_1 : s > 30.$
No. Hypotheses are always statements about the population or distribution under study, not statements about the sample.

Example 1.2 (Semiconductor tool)**.** A semiconductor manufacturer collects data from a new tool and conducts a hypothesis test with the null hypothesis that a critical dimension mean width equals 100 nm. The conclusion is to reject the null hypothesis. Does this result provide strong evidence that the critical-dimension mean equals 100 nm? Explain.

Solution. The truth or falsity of a particular hypothesis can never be known with certainty unless we can examine the entire population. This is usually impossible in most practical situations. Therefore, a hypothesis-testing procedure should be developed with the probability of reaching a wrong conclusion in mind. Rejecting H_0 is a *strong* conclusion, so the result provides strong evidence *against* the mean equalling 100 nm.

Example 1.3 (Connector pull-off force)**.** The mean pull-off force of a connector depends on cure time.

(a) State the null and alternative hypotheses used to demonstrate that the pull-off force is below 25 N.

(b) Assume that the previous test does not reject the null hypothesis. Does this result provide strong evidence that the pull-off force is greater than or equal to 25 N? Explain.

Solution.

(a) The null and alternative hypotheses used to demonstrate that the pull-off force is below 25 N are

$$H_0 : \mu = 25\,\text{N}, \qquad H_1 : \mu < 25\,\text{N}.$$

(b) **No.** The truth or falsity of a particular hypothesis can never be known with certainty unless we can examine the entire population. Failing to reject H_0 is a weak conclusion and does not provide strong evidence that the pull-off force is ≥ 25 N.

Chapter 2

Tests on the Mean of a Normal Distribution, Variance Known

We now specialise the general framework of Chapter 1 to the most fundamental case: testing the mean μ of a normal population when the variance σ^2 is known. This chapter first works through several detailed error-probability calculations, then states the general theory (test statistic, rejection criteria, P-values, Type II error and sample size), and closes with a set of worked problems.

2.1 Computing α and β: Worked Examples

Example 2.1 (Textile fibre — yarn elongation)**.** A textile fibre manufacturer is investigating a new drapery yarn, which the company claims has a mean thread elongation of 12 kg with a standard deviation of 0.5 kg. The company wishes to test the hypothesis $H_0 : \mu = 12$ against $H_1 : \mu < 12$, using a random sample of $n = 4$ specimens.

(a) What is the Type I error probability if the critical region is defined as $\bar{x} < 11.5$ kg?

(b) Find β for the case where the true mean elongation is 11.25 kg.

(c) Find β for the case where the true mean is 11.5 kg.

Solution. **(a)** The probability of making a Type I error is

$$\alpha = \mathrm{P}(\text{Type I error}) = \mathrm{P}(\text{reject } H_0 \mid H_0 \text{ true}).$$

Here $\mu = 12$ kg, $\sigma = 0.5$ kg, and the critical region is $\bar{x} < 11.5$ (reject H_0).

$$\alpha = \mathrm{P}(\bar{x} < 11.5 \mid \mu = 12) = \mathrm{P}\left(\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} < \frac{11.5 - \mu}{\sigma/\sqrt{n}} \;\middle|\; \mu = 12 \right)$$

$$= \mathrm{P}\left(Z < \frac{11.5 - 12}{0.5/\sqrt{4}} \right) = \mathrm{P}(Z < -2).$$

From the standard normal tables, for $Z < -2$, the probability is 0.022750.

$$\boxed{\alpha = 0.02275.}$$

(b) The probability of a Type II error is $\beta = \mathrm{P}(\text{fail to reject } H_0 \mid H_0 \text{ false})$. With the true mean $\mu = 11.25$,

$$\beta = \mathrm{P}(\bar{x} > 11.5 \mid \mu = 11.25) = \mathrm{P}\left(\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} > \frac{11.5 - 11.25}{0.5/\sqrt{4}}\right)$$

$$= \mathrm{P}(Z > 1) = 1 - \mathrm{P}(Z \leq 1) = 1 - 0.841345 = 0.15865.$$

$$\boxed{\beta = 0.15865.}$$

(c) With the true mean $\mu = 11.5$,

$$\beta = \mathrm{P}(\bar{x} > 11.5 \mid \mu = 11.5) = \mathrm{P}\left(Z > \frac{11.5 - 11.5}{0.5/\sqrt{4}}\right) = \mathrm{P}(Z > 0) = 1 - \mathrm{P}(Z \leq 0) = 1 - 0.5.$$

$$\boxed{\beta = 0.5.}$$

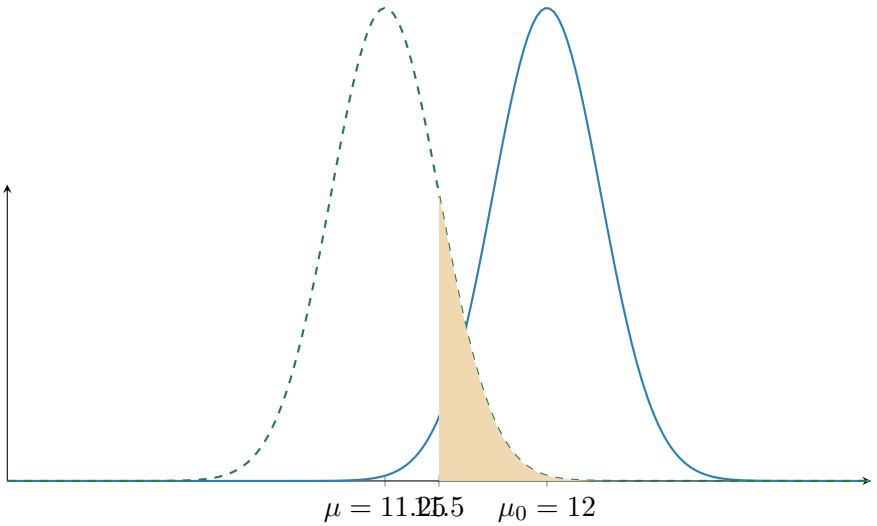


Figure 2.1: Type II error (β, shaded) when the true mean is 11.25: the null curve is centred at $\mu_0 = 12$ and the true curve at $\mu = 11.25$.

Example 2.2 (Specialty parts — customer inspection)**.** The length L of a certain specialty part is normally distributed with a variance of $50\,\mathrm{mm}^2$. A customer wants the mean length to be 100 mm but measures the average length in a randomly selected sample as 103 mm. What is the conclusion about the sample? The sample size is $n = 50$ parts.

Solution.

$$H_0 : \mu = 100 \qquad \text{vs.} \qquad H_1 : \mu \neq 100.$$

The test statistic is

$$Z_0 = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{103 - 100}{\sqrt{50/50}} = \frac{3}{1} = 3.$$

Taking a level of confidence of 95% and assuming $\alpha = 5\%$,

$$Z_{\alpha/2} = Z_{0.025} = 1.96.$$

Since $Z_0 = 3 > Z_{\alpha/2} = 1.96$, we reject H_0. Hence the mean length of the parts $\neq 100$, and **the customer rejects the sample.**

Example 2.3 (Yarn elongation continued — effect of sample size)**.** Reconsider the textile-fibre problem, but now with $n = 16$ and the same critical region $\bar{x} < 11.5$.

Solution.

1. Type I error:

$$\alpha = \mathrm{P}(\bar{x} < 11.5 \mid \mu = 12) = \mathrm{P}\left(Z < \frac{11.5 - 12}{0.5/\sqrt{16}}\right) = \mathrm{P}\left(Z < -\tfrac{0.5}{0.5/4}\right) = \mathrm{P}(Z < -4) = 0.000.$$

2. Type II error at $\mu = 11.25$:

$$\beta = \mathrm{P}(\bar{x} > 11.5 \mid \mu = 11.25) = \mathrm{P}\left(Z > \frac{11.5 - 11.25}{0.5/\sqrt{16}}\right) = \mathrm{P}\left(Z > \tfrac{0.25}{0.5/4}\right)$$

$$= \mathrm{P}(Z > 2) = 1 - \mathrm{P}(Z \le 2) = 0.022750.$$

3. Type II error at $\mu = 11.5$:

$$\beta = \mathrm{P}(\bar{x} > 11.5 \mid \mu = 11.5) = \mathrm{P}\left(Z > \frac{11.5 - 11.5}{0.5/\sqrt{16}}\right) = \mathrm{P}(Z > 0) = 1 - \mathrm{P}(Z \le 0) = 0.5.$$

Increasing the sample size from 4 to 16 has sharply reduced both α (from 0.02275 to ≈ 0) and β at $\mu = 11.25$ (from 0.15865 to 0.02275), while β at $\mu = 11.5$ (where H_0 is barely false) remains 0.5.

Example 2.4 (Boundary of the critical region)**.** For the yarn problem, find the boundary C of the critical region $\bar{x} < C$ if the Type I error probability is

(a) $\alpha = 0.01$ and $n = 4$;

(b) $\alpha = 0.05$ and $n = 4$;

(c) $\alpha = 0.01$ and $n = 16$.

Solution. **(a)** $\alpha = 0.01$, $n = 4$:

$$0.01 = \mathrm{P}\left(Z \le \frac{C - 12}{0.5/\sqrt{4}}\right).$$

A probability of 0.01 corresponds to $Z = -2.33$, so

$$\frac{C - 12}{0.5/\sqrt{4}} = -2.33 \;\Rightarrow\; C = 11.4175.$$

(b) $\alpha = 0.05$, $n = 4$: $Z = -1.65$, so

$$\frac{C - 12}{0.5/\sqrt{4}} = -1.65 \;\Rightarrow\; C = 11.5875.$$

(c) $\alpha = 0.01$, $n = 16$: $Z = -2.33$, so

$$\frac{C - 12}{0.5/\sqrt{16}} = -2.33 \;\Rightarrow\; C = -0.29125 + 12 = 11.70875.$$

Example 2.5 (Cement — heat evolved)**.** The heat evolved in calories per gram of a cement mixture is approximately normally distributed. The mean is thought to be 100 and the standard deviation is 2. We wish to test $H_0 : \mu = 100$ versus $H_1 : \mu \neq 100$ with a sample of $n = 9$ specimens.

(a) If the acceptance region is defined as $98.5 \leq \bar{x} \leq 101.5$, find the Type I error probability.

(b) Find β for the case where the true mean heat evolved is 103.

(c) Find β for the case where the true mean heat evolved is 105. Compare β in (b) and (c) and state reasons.

Solution. Here $H_0 : \mu = 100$, $H_1 : \mu \neq 100$, $n = 9$.

(a) P(Type I error) $= \alpha$, with acceptance region $98.5 \leq \bar{x} \leq 101.5$:

$$\alpha = \mathrm{P}(\bar{x} < 98.5) + \mathrm{P}(\bar{x} > 101.5)$$

$$= \mathrm{P}\left(Z < \frac{98.5 - 100}{2/\sqrt{9}}\right) + \mathrm{P}\left(Z > \frac{101.5 - 100}{2/\sqrt{9}}\right) = \mathrm{P}\left(Z < \frac{(-1.5) \cdot 3}{2}\right) + \mathrm{P}\left(Z > \frac{4.5}{2}\right)$$

$$= 0.012224 + \big[1 - \mathrm{P}(Z < 2.25)\big] = 0.012224 + 1 - 0.987776 = 0.024448.$$

(b) At $\mu = 103$:

$$\beta = \mathrm{P}(98.5 \leq \bar{x} \leq 101.5 \mid \mu = 103) = \mathrm{P}\left(\frac{98.5 - \mu}{\sigma/\sqrt{n}} \leq Z \leq \frac{101.5 - \mu}{\sigma/\sqrt{n}}\right)$$

$$= \mathrm{P}\left(\frac{98.5 - 103}{2/\sqrt{9}} \leq Z \leq \frac{101.5 - 103}{2/\sqrt{9}}\right) = \mathrm{P}(-6.75 \leq Z \leq -2.25)$$

$$= \mathrm{P}(Z \leq -2.25) - \mathrm{P}(Z \leq -6.75).$$

$$\boxed{\beta = 0.012224.}$$

(c) At $\mu = 105$:

$$\beta = \mathrm{P}\left(\frac{98.5 - 105}{2/\sqrt{9}} \leq Z \leq \frac{101.5 - 105}{2/\sqrt{9}}\right) = \mathrm{P}\left(Z < \frac{101.5 - 105}{2/3}\right) - \mathrm{P}\left(Z \leq \frac{98.5 - 105}{2/3}\right)$$

$$= \mathrm{P}(Z < -5.25) - \mathrm{P}(Z < -9.75) = 0 - 0.$$

$$\boxed{\beta = 0.}$$

We observe that when the true mean is 103 there is about a 1% chance of failing to reject the null hypothesis even though it is false, but when the true mean is 105 we will definitely reject the null hypothesis for the given confidence interval. The larger the departure of the true mean from the hypothesized value, the smaller β becomes.

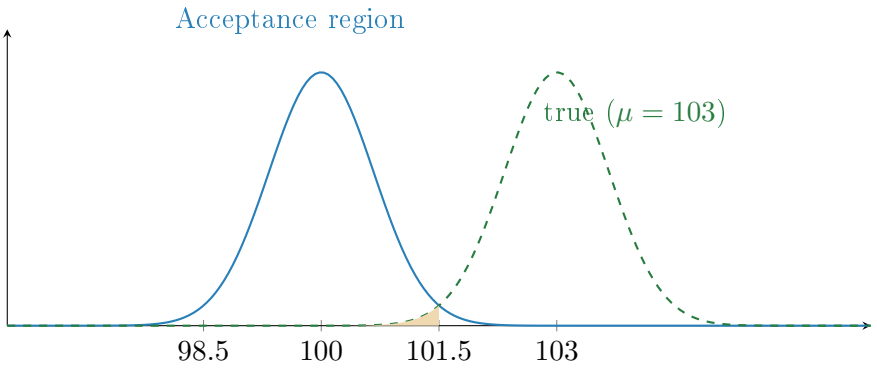


Figure 2.2: Type II error (β, shaded) when the true mean is 103; the acceptance region $[98.5, 101.5]$ is set under the null curve centred at 100.

Example 2.6 (Power supply output voltage)**.** A manufacturer is interested in the output voltage of a power supply used in a PC. Output voltage is assumed to be normally distributed with standard deviation 0.25 V, and the manufacturer wishes to test $H_0 : \mu = 5\,\text{V}$ against $H_1 : \mu \neq 5\,\text{V}$ using $n = 8$ units.

(a) The acceptance region is $4.85 \leq \bar{x} \leq 5.15$. Find the value of α.

(b) Find the power of the test for detecting a true mean output voltage of 5.1 V.

Solution. $H_0 : \mu = 5$ vs. $H_1 : \mu \neq 5$, $n = 8$, $\sigma = 0.25$.

(a)

$$\alpha = \text{P}\left(\frac{4.85 - \mu}{0.25/\sqrt{8}} \leq Z \leq \frac{5.15 - \mu}{0.25/\sqrt{8}}\right)^{c}$$

Evaluated at the boundaries with $\mu = 5$:

$$\alpha = \text{P}(Z \leq -1.70) + \big(1 - \text{P}(Z \leq 1.70)\big) = 0.044565 + (1 - 0.955435)$$

$$= 0.044565 + 0.044565 = 0.089130 \approx 0.089.$$

$$\boxed{\alpha = 0.089130.}$$

(b) Power $= 1 - \beta$ at $\mu = 5.1\,\text{V}$:

$$\beta = \text{P}\left(\frac{4.85 - \mu}{0.25/\sqrt{8}} \leq Z \leq \frac{5.15 - \mu}{0.25/\sqrt{8}} \,\middle|\, \mu = 5.1\right) = \text{P}\left(\frac{4.85 - 5.1}{0.25/\sqrt{8}} \leq Z \leq \frac{5.15 - 5.1}{0.25/\sqrt{8}}\right)$$

$$= \text{P}(-2.8284 \leq Z \leq 0.5657) = \text{P}(Z < 0.5657) - \text{P}(Z < -2.8284) = 0.71186.$$

$$\therefore\ \text{Power} = 1 - \beta = 1 - 0.71186 = \boxed{0.28814.}$$

Example 2.7 (Phoenix toll roads — a proportion via the binomial)**.** The proportion of residents in Phoenix favouring the building of toll roads to complete the freeway system is believed to be $p = 0.3$. If a random sample of 10 residents shows that 1 or fewer favour this proposal, we will conclude $p < 0.3$.

(a) Find the probability of a Type I error if the true proportion is $p = 0.3$.

(b) Find the probability of committing a Type II error with this procedure if $p = 0.2$.

(c) What is the power of this procedure if the true proportion is $p = 0.2$?

Solution. $H_0 : p = 0.3$, $H_1 : p < 0.3$, $n = 10$. Let X be the number of residents in favour of the proposal. Then X is a binomial random variable with parameters $n = 10$ and p:

$$\mathrm{P}(X = x) = \binom{n}{x} p^x (1-p)^{10-x}, \qquad x \in \{0, 1, \ldots, n\}.$$

(a)

$$\alpha = \mathrm{P}(X \le 1 \mid p = 0.3) = \mathrm{P}(X = 0 \mid p = 0.3) + \mathrm{P}(X = 1 \mid p = 0.3)$$
$$= \binom{10}{0} 0.3^0 (1-0.3)^{10} + \binom{10}{1} 0.3^1 (1-0.3)^9 = 1 \cdot 0.7^{10} + 10 \cdot 0.3 \cdot 0.7^9.$$
$$\boxed{\alpha = 0.1493.}$$

(b)

$$\beta = \mathrm{P}(X > 1 \mid p = 0.2) = 1 - \mathrm{P}(X \le 1 \mid p = 0.2) = 1 - \big[\mathrm{P}(X = 0) + \mathrm{P}(X = 1)\big]_{p=0.2}$$
$$= 1 - \left[\binom{10}{0} 0.2^0 (1-0.2)^{10} + \binom{10}{1} 0.2^1 (1-0.2)^9\right] = 1 - \left[1 \cdot 0.8^{10} + 10 \cdot 0.2 \cdot 0.8^9\right].$$
$$\boxed{\beta = 0.6242.}$$

(c) If the true proportion is $p = 0.2$, the power is

$$1 - \beta = 1 - 0.6242 = \boxed{0.3758.}$$

2.2 General Theory: Test on the Mean, Variance Known

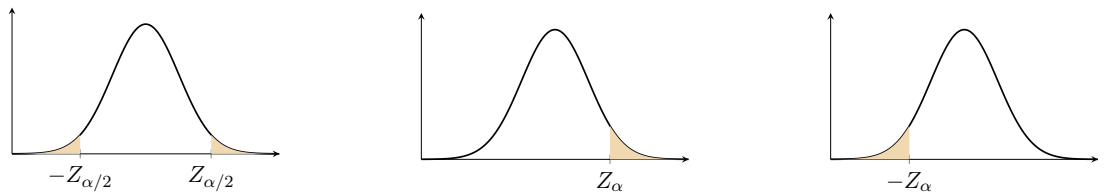
Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal population with known variance σ^2, and let μ_0 be a specified constant. The hypotheses are

$$H_0 : \mu = \mu_0, \qquad H_1 : \mu \ne \mu_0 \quad \text{or} \quad H_1 : \mu > \mu_0 \quad \text{or} \quad H_1 : \mu < \mu_0.$$

Test Statistic (σ known)

$$Z_0 = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}.$$

Under H_0, Z_0 follows the standard normal distribution $N(0, 1)$.



(a) $H_1 : \mu \ne \mu_0$ (b) $H_1 : \mu > \mu_0$ (c) $H_1 : \mu < \mu_0$

Figure 2.3: The distribution of Z_0 when $H_0 : \mu = \mu_0$ is true, with critical regions (shaded) for (a) the two-sided alternative, (b) the upper one-sided alternative, and (c) the lower one-sided alternative.

Two-sided test

For a two-sided test we reject H_0 if the observed value of the test statistic Z_0 is either

$$Z_0 > Z_{\alpha/2} \quad \text{or} \quad Z_0 < -Z_{\alpha/2}, \tag{1}$$

and we fail to reject H_0 if

$$-Z_{\alpha/2} \le Z_0 \le Z_{\alpha/2}. \tag{2}$$

The two inequalities in (1) define the critical (rejection) region; equation (2) defines the acceptance region. Equivalently, this can be written as:

Two-sided rejection in terms of $\bar{x}$

Reject $H_0 : \mu = \mu_0$ if either $\bar{x} > a$ or $\bar{x} < b$, where

$$a = \mu_0 + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \qquad b = \mu_0 - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

One-sided alternatives

For $H_0 : \mu = \mu_0$ vs. $H_1 : \mu > \mu_0$, the rejection criterion is

$$Z_0 > Z_\alpha \qquad (\text{equivalently } \bar{x} > \mu_0 + Z_\alpha\,\sigma/\sqrt{n}).$$

For $H_0 : \mu = \mu_0$ vs. $H_1 : \mu < \mu_0$, the rejection criterion is

$$Z_0 < -Z_\alpha.$$

2.3 P-values for the Normal Test

P-value (σ known)

$$P = \begin{cases} 2\big[1 - \Phi(|Z_0|)\big], & \text{two-tailed test } (H_1 : \mu \neq \mu_0), \\ 1 - \Phi(Z_0), & \text{upper-tailed test } (H_1 : \mu > \mu_0), \\ \Phi(Z_0), & \text{lower-tailed test } (H_1 : \mu < \mu_0), \end{cases}$$

where $\Phi(z) = \mathrm{P}(Z \le z)$ is the standard normal cumulative distribution function, $Z \sim N(0,1)$.

The P-value is the smallest level of significance that would lead to rejection of the null hypothesis H_0 with the given data.

Type II error for a two-sided test (σ known)

$$\beta = \Phi\left(Z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right) - \Phi\left(-Z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right),$$

and if $\delta > 0$, approximately $\beta \approx \Phi\left(Z_{\alpha/2} - \dfrac{\delta\sqrt{n}}{\sigma}\right)$, where $\delta = \mu - \mu_0$.

Sample size

$$n \approx \frac{(Z_{\alpha/2} + Z_\beta)^2 \, \sigma^2}{\delta^2}, \qquad \delta = \mu - \mu_0.$$

2.4 Worked Problems

Example 2.8 (Stating hypotheses)**.** State the null and alternative hypotheses in each case.

(a) A hypothesis test will be used to potentially provide evidence that the population mean is greater than 10.

(b) A hypothesis test will be used to potentially provide evidence that the population mean is not equal to 7.

(c) A hypothesis test will be used to potentially provide evidence that the population mean is less than 5.

Solution.

(a) $H_0 : \mu = 10,\ H_1 : \mu > 10$; (b) $H_0 : \mu = 7,\ H_1 : \mu \neq 7$; (c) $H_0 : \mu = 5,\ H_1 : \mu < 5$.

Example 2.9 (Critical values for an upper one-sided test)**.** A hypothesis will be used to test that a population mean equals 10 against the alternative that the population mean is greater than 10, with known variance σ. What are the critical values for the test statistic Z_0 for the following significance levels: (a) 0.01, (b) 0.05, (c) 0.10?

Solution. $H_0 : \mu = 10$ vs. $H_1 : \mu > 10$, test statistic $Z_0 = (\bar{x} - \mu)/(\sigma/\sqrt{n})$. For a one-sided alternative, $P = 1 - \Phi(Z_c)$.

(a) $\alpha = 0.01$: $\Phi(Z_c) = 1 - 0.01 = 0.99 \Rightarrow Z_c = 2.33$.

(b) $\alpha = 0.05$: $\Phi(Z_c) = 1 - 0.05 = 0.95 \Rightarrow Z_c = 1.65$.

(c) $\alpha = 0.10$: $\Phi(Z_c) = 1 - 0.10 = 0.90 \Rightarrow Z_c = 1.29$.

Example 2.10 (Critical values for a two-sided test)**.** A hypothesis will be used to test that a population mean equals 7 against the alternative that the population mean does not equal 7, with known variance σ. What are the critical values for the test statistic Z_0 for significance levels (a) 0.01, (b) 0.05, (c) 0.10?

Solution. $H_0 : \mu = 7$ vs. $H_1 : \mu \neq 7$; P-value $= 2\big[1 - \Phi(|Z_c|)\big]$.

(a) $\alpha = 0.01$: $\mathrm{P}(Z_0 < |Z_c|) = 1 - \frac{0.01}{2} = 0.995 \Rightarrow Z_c = 2.58 \Rightarrow Z_c = \pm 2.58$.

(b) $\alpha = 0.05$: $\mathrm{P}(Z_0 < |Z_c|) = 1 - \frac{0.05}{2} = 0.975 \Rightarrow Z_c = 1.96 \Rightarrow Z_c = \pm 1.96$.

(c) $\alpha = 0.10$: $\mathrm{P}(Z_0 < |Z_c|) = 1 - \frac{0.10}{2} = 0.95 \Rightarrow Z_c = \pm 1.65$.

Example 2.11 (P-values for a two-sided test)**.** For the hypothesis test $H_0 : \mu = 7$ and $H_1 : \mu \neq 7$ with variance known, calculate the P-value for each of the following test statistics: (a) $Z_0 = 2.05$, (b) $Z_0 = -1.84$, (c) $Z_0 = 0.4$.

Solution. $P = 2\big[1 - \Phi(|Z_0|)\big]$.

(a) $Z_0 = 2.05$: $P = 2\big[1 - \Phi(2.05)\big] = 2\big[1 - 0.979818\big] = 2(0.020182) = 0.040364.$

(b) $Z_0 = -1.84$: $P = 2\big[1 - \Phi(1.84)\big] = 2\big[1 - 0.967116\big] = 2(0.032884) = 0.064668$ (i.e. $0.6466\ldots$ per the notes' arithmetic; see remark).

(c) $Z_0 = 0.4$: $P = 2\big[1 - \Phi(0.4)\big] = 2\big[1 - 0.655422\big] = 2(0.344578) = 0.689156 \approx 0.69.$

Example 2.12 (Crankshaft wear)**.** A manufacturer produces crankshafts for an automobile engine. The wear of the crankshaft after 100 000 miles (0.0001 inch) is of interest because it is likely to have an impact on warranty claims. A random sample of $n = 15$ shafts is tested and $\bar{x} = 2.78$. It is known that $\sigma = 0.9$ and that wear is normally distributed.

(a) Test $H_0 : \mu = 3$ versus $H_1 : \mu \neq 3$ using $\alpha = 0.05$.

(b) What is the power of this test if $\mu = 3.25$?

(c) What sample size would be required to detect a true mean of 3.75 if we wanted the power to be at least 0.9?

Solution. $H_0 : \mu = 3$ vs. $H_1 : \mu \neq 3$, $\alpha = 0.05$, $\bar{x} = 2.78$, $\sigma = 0.9$, $n = 15$.

(a)

$$Z_0 = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{2.78 - 3}{0.9/\sqrt{15}} = -0.9467.$$

Two-sided test, $Z_{\alpha/2} = Z_{0.025} = 1.96 = \pm 1.96$. Rejection criterion: $Z_0 > Z_{\alpha/2}$ or $Z_0 < -Z_{\alpha/2}$. Here $Z_0 = -0.95 > -1.96$ but $-0.95 < 1.96$, so **we fail to reject H_0.**

(b) With $\delta = \mu - \mu_0 = 3.25 - 3 = 0.25$,

$$\beta = \Phi\left(Z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right) - \Phi\left(-Z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right)$$

$$= \Phi\left(1.96 - \frac{0.25\sqrt{15}}{0.9}\right) - \Phi\left(-1.96 - \frac{0.25\sqrt{15}}{0.9}\right) = \Phi(0.88) - \Phi(-3.04) = 0.809387.$$

$$\therefore \text{ Power} = 1 - \beta = 1 - 0.80939 = \boxed{0.19061.}$$

(c) Sample size with $\mu = 3.75$, $\delta = 3.75 - 3.0 = 0.75$, power $= 0.9 \Rightarrow \beta = 0.1$, $Z_\beta = Z_{0.1} = 1.28$:

$$n = \frac{(Z_{\alpha/2} + Z_\beta)^2 \sigma^2}{\delta^2} = \frac{(1.96 + 1.28)^2 (0.9)^2}{(0.75)^2} = 15.116.$$

$$\boxed{n \approx 16.}$$

Example 2.13 (Battery life)**.** The life in hours of a battery is known to be approximately normally distributed with standard deviation $\sigma = 1.25\,\text{h}$. A random sample of 10 batteries has a mean life of $\bar{x} = 40.5\,\text{h}$.

(a) Is there evidence to support the claim that battery life exceeds 40 h? Use $\alpha = 0.05$.

(b) What is the P-value for the test in part (a)?

(c) What is the β-error for the test in part (a) if the true mean life is 42 h?

(d) What sample size would be required to ensure that β does not exceed 0.10 if the true mean life is 44 h?

(e) Explain how you could answer the question in part (a) by calculating an appropriate confidence bound on life.

Solution. $H_0 : \mu = 40$ vs. $H_1 : \mu > 40$, $\alpha = 0.05$, $n = 10$, $\sigma = 1.25$.

(a)

$$Z_0 = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{40.5 - 40}{1.25/\sqrt{10}} = 1.26, \qquad Z_\alpha = Z_{0.05} = 1.65.$$

For a one-sided test $\mu > \mu_0$, the rejection criterion is $Z_0 > Z_\alpha$. Here $Z_0 = 1.26$ and $Z_\alpha = 1.65$; since $Z_0 < Z_\alpha$, **we fail to reject** H_0. There is insufficient evidence to support the claim that battery life exceeds 40 h.

(b)

$$P = \mathrm{P}(Z > Z_0) = 1 - \Phi(Z_0) = 1 - \Phi(1.26) = 1 - 0.896165 = 0.103835.$$

(c) With $\delta = \mu - \mu_0 = 42 - 40 = 2$,

$$\beta = \Phi\left(Z_\alpha - \frac{\delta\sqrt{n}}{\sigma}\right) = \Phi\left(1.65 - \frac{2\sqrt{10}}{1.25}\right) = \Phi(-3.41).$$

$$\boxed{\beta = 0.000325.}$$

(d) With $\mu = 44$, $\delta = 4$, $\beta < 0.1$, $Z_\beta = Z_{0.1} = 1.29$:

$$n = \frac{(Z_\alpha + Z_\beta)^2\sigma^2}{\delta^2} = \frac{(1.65 + 1.29)^2(1.25)^2}{4^2} = 0.844.$$

$$\boxed{n \approx 1.}$$

(e) A $100(1-\alpha)\%$ lower confidence bound on the mean is

$$\bar{x} - Z_\alpha \frac{\sigma}{\sqrt{n}} \le \mu.$$

Thus the 95% lower bound on the mean is

$$40.5 - 1.65 \cdot \frac{1.25}{\sqrt{10}} \le \mu \;\Rightarrow\; 39.85 \le \mu.$$

Since $40 \in (39.85,\ \mu)$, the bound provides evidence to support that battery life ≈ 40 h (consistent with failing to reject H_0 in part a).

Example 2.14 (Tensile strength of a steel alloy)**.** An engineer studying the tensile strength of a steel alloy intended for golf-club shafts knows that tensile strength is approximately normally distributed with $\sigma = 60$ psi. A random sample of 12 specimens has mean tensile strength $\bar{x} = 3450$ psi.

(a) Test the hypothesis that mean strength is 3500 psi. Use $\alpha = 0.01$.

(b) What is the smallest level of significance at which you would be willing to reject the null hypothesis?

(c) What is the β-error for the test in part (a) if the true mean is 3470?

(d) Suppose we wanted to reject the null hypothesis with probability at least 0.8 if mean strength $\mu = 3500$. What sample size should be used?

(e) Explain how you could answer the question in part (a) with a two-sided confidence interval on mean tensile strength.

Solution. Given $n = 12$, $\bar{x} = 3450$, $\sigma = 60$, $H_0 : \mu = 3500$, $H_1 : \mu \neq 3500$.

(a)

$$Z_0 = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{3450 - 3500}{60/\sqrt{12}} = -2.89.$$

$\alpha = 0.01 \Rightarrow Z_{\alpha/2} = Z_{0.005} = \pm 2.58$. Rejection criterion: $Z_0 < -Z_{\alpha/2}$ or $Z_0 > Z_{\alpha/2}$. Since $Z_0 = -2.89 < -2.58$, we **reject the null hypothesis**, i.e. $\mu = 3500$ is not true.

(b) Smallest level of significance (P-value):

$$P = 2\big[1 - \Phi(|Z_0|)\big] = 2\big[1 - \Phi(2.89)\big] = 2\big[1 - 0.998074\big] = 0.003852.$$

(c) With $\mu_0 = 3500$, $\mu = 3470$, $\delta = \mu - \mu_0 = -30$:

$$\beta = \Phi\left(Z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right) - \Phi\left(-Z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right)$$

$$= \Phi\left(2.58 - \frac{-30\sqrt{12}}{60}\right) - \Phi\left(-2.58 - \frac{-30\sqrt{12}}{60}\right) = \Phi(4.31) - \Phi(-0.85)$$

$$= 1 - 0.197662 = \boxed{0.802338.}$$

(d) Sample size with $\mu = 3470$, $\delta = -30$, power $= 0.8 \Rightarrow \beta = 0.2$, $Z_\beta = Z_{0.2} = 0.84$:

$$n = \frac{(Z_{\alpha/2} + Z_\beta)^2\sigma^2}{\delta^2} = \frac{(2.58 + 0.84)^2 \cdot 60^2}{(-30)^2} = 46.78.$$

$$\boxed{n \approx 47.}$$

(e) A $100(1-\alpha)\%$ two-sided confidence interval on the mean is

$$\bar{x} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

$$3450 - 2.58 \cdot \frac{60}{\sqrt{12}} \leq \mu \leq 3450 + 2.58 \cdot \frac{60}{\sqrt{12}} \;\Rightarrow\; 3405.31 \leq \mu \leq 3494.67.$$

As 3500 is not in the interval, we reject H_0, i.e. $\mu = 3500$ is not true for $\alpha = 0.01$.

Example 2.15 (Bearing inside diameter)**.** A bearing used in an automotive application is supposed to have a nominal inside diameter of 1.5 inches. A random sample of 25 bearings is selected, and the average inside diameter is 1.4975 inches. Bearing diameter is known to be normally distributed with $\sigma = 0.01$ inch.

(a) Test $H_0 : \mu = 1.5$ vs. $H_1 : \mu \neq 1.5$ using $\alpha = 0.01$.

(b) What is the P-value for the test?

(c) Compute the power of the test if the true mean diameter is 1.495 inches.

(d) What sample size would be required to detect a true mean diameter as low as $1.495''$ if we wanted the power of the test to be at least 0.9?

(e) Explain how part (a) can be answered by constructing a two-sided confidence interval on the mean diameter.

Solution. $n = 25$, $\bar{x} = 1.4975''$, $\sigma = 0.01''$, $H_0 : \mu = 1.5$, $H_1 : \mu \neq 1.5$, $\alpha = 0.01$.

(a)

$$Z_0 = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{1.4975 - 1.5}{0.01/\sqrt{25}} = -1.25, \qquad Z_{\alpha/2} = Z_{0.005} = \pm 2.58.$$

Rejection criterion: $Z_0 < -Z_{\alpha/2}$ or $Z_0 > Z_{\alpha/2}$. Since $Z_0 = -1.25 > -2.58$ and $-1.25 < 2.58$, **we fail to reject** H_0; hence $H_0 : \mu = 1.5$ is (taken to be) true.

(b)

$$P = 2\big[1 - \Phi(|Z_0|)\big] = 2\big[1 - \Phi(1.25)\big] = 2\big[1 - 0.894350\big] = 0.2113.$$

(c) Power $= 1 - \beta$. With $\delta = \mu - \mu_0 = 1.495 - 1.5 = -0.005$,

$$\beta = \Phi\left(Z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right) - \Phi\left(-Z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right)$$

$$= \Phi\left(2.58 - \frac{-0.005\sqrt{25}}{0.01}\right) - \Phi\left(-2.58 - \frac{-0.005\sqrt{25}}{0.01}\right)$$

$$= \Phi(5.08) - \Phi(-0.08) = 1 - 0.468119 = 0.531881.$$

$$\therefore \text{ Power} = 1 - \beta = 1 - 0.531881 = \boxed{0.468119.}$$

(d) Sample size with $\delta = 1.495 - 1.5 = -0.005$, power $= 0.9 \Rightarrow \beta = 0.1$, $Z_{0.1} = 1.28$:

$$n = \frac{(Z_{\alpha/2} + Z_\beta)^2\sigma^2}{\delta^2} = \frac{(2.58 + 1.28)^2 \cdot 0.01^2}{(-0.005)^2} = 59.6.$$

$$\boxed{n \approx 60.}$$

(e) A $100(1-\alpha)\%$ two-sided confidence interval on the mean:

$$\bar{x} - Z_{\alpha/2}\,\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + Z_{\alpha/2}\,\frac{\sigma}{\sqrt{n}}$$

$$1.4975 - 2.58 \cdot \frac{0.01}{\sqrt{25}} \leq \mu \leq 1.4975 + 2.58 \cdot \frac{0.01}{\sqrt{25}} \;\Rightarrow\; 1.49234 \leq \mu \leq 1.50266.$$

Since 1.5 is in the interval $[1.49234,\ 1.50266]$, we conclude that $H_0 : \mu = 1.5$ is true and cannot be rejected.

Chapter 3

Tests on the Variance and Standard Deviation of a Normal Distribution

Many engineering problems concern the *variability* of a process rather than its mean. This chapter develops the chi-square test for the variance σ^2 (equivalently the standard deviation σ) of a normal distribution, together with the associated confidence bounds.

3.1 Hypothesis Tests on the Variance

The hypotheses are

$$H_0 : \sigma^2 = \sigma_0^2, \qquad H_1 : \sigma^2 \neq \sigma_0^2.$$

Chi-square Test Statistic

$$\chi_0^2 = \frac{(n-1)s^2}{\sigma_0^2}.$$

Under H_0, χ_0^2 follows a chi-square distribution with $n-1$ degrees of freedom.

Rejection Criteria

For the two-sided alternative $H_1 : \sigma^2 \neq \sigma_0^2$, reject H_0 if

$$\chi_0^2 > \chi^2_{\alpha/2,\,n-1} \quad \text{or} \quad \chi_0^2 < \chi^2_{1-\alpha/2,\,n-1}.$$

For the one-sided alternatives:

$$H_1 : \sigma^2 > \sigma_0^2 : \quad \chi_0^2 > \chi^2_{\alpha,\,n-1}; \qquad H_1 : \sigma^2 < \sigma_0^2 : \quad \chi_0^2 < \chi^2_{1-\alpha,\,n-1}.$$

Table 3.1: Rejection criteria for tests on the variance.

Alternative hypothesis	Rejection criterion
$H_1 : \sigma^2 \neq \sigma_0^2$	$\chi_0^2 > \chi^2_{\alpha/2,n-1}$ or $\chi_0^2 < \chi^2_{1-\alpha/2,n-1}$
$H_1 : \sigma^2 > \sigma_0^2$	$\chi_0^2 > \chi^2_{\alpha,n-1}$
$H_1 : \sigma^2 < \sigma_0^2$	$\chi_0^2 < \chi^2_{1-\alpha,n-1}$

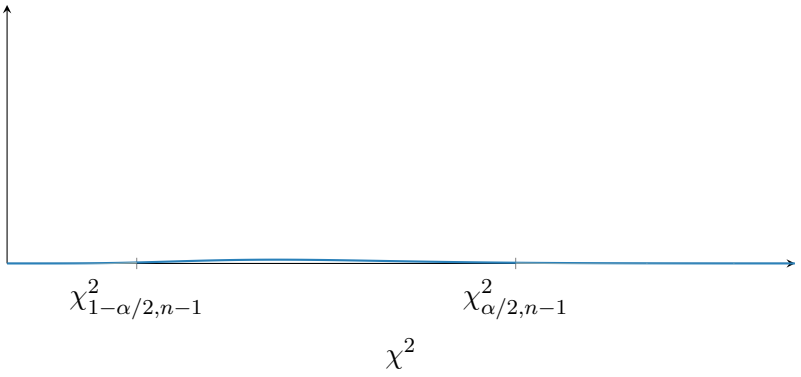


Figure 3.1: Chi-square distribution with $n-1$ degrees of freedom, showing the two-sided critical regions (shaded).

3.2 Worked Problems

Example 3.1 (Detergent filling machine)**.** An automated filling machine is used to fill bottles with liquid detergent. A random sample of 20 bottles results in a sample variance of fill volume of $s^2 = 0.0153$ (fluid ounces)^2. If the variance of fill volume exceeds 0.01 (fluid ounces)^2, an unacceptable proportion of bottles will be underfilled or overfilled. Is there evidence in the sample data to suggest that the manufacturer has a problem with underfilled or overfilled bottles? Use $\alpha = 0.05$ and assume that fill volume has a normal distribution.

Solution (eight-step procedure).

1. The parameter of interest is the population variance σ^2.
2. $H_0 : \sigma^2 = 0.01$.
3. $H_1 : \sigma^2 > 0.01$.
4. $\alpha = 0.05$.
5. Test statistic: $\chi^2_0 = \dfrac{(n-1)s^2}{\sigma^2_0}$.
6. Reject H_0 if $\chi^2_0 > \chi^2_{0.05,19} = 30.14$.
7. Computation:
$$\chi^2_0 = \frac{19 \times 0.0153}{0.01} = 29.07.$$
8. Conclusion: Since $\chi^2_0 = 29.07 < \chi^2_{0.05,19} = 30.14$, we conclude that there is no strong evidence that the variance of fill volume exceeds 0.01 (fluid ounces)^2.

Example 3.2 (Critical values, two-sided variance test)**.** Consider the test of $H_0 : \sigma^2 = 7$ against $H_1 : \sigma^2 \neq 7$. What are the critical values for the test statistic χ^2_0 for the following significance levels and sample sizes?

(a) $\alpha = 0.01$ and $n = 20$;

(b) $\alpha = 0.05$ and $n = 12$;

(c) $\alpha = 0.10$ and $n = 15$.

Solution.

(a) $\alpha = 0.01$, $n = 20$: $\alpha/2 = 0.005$, $1 - \alpha/2 = 0.995$; for $n - 1 = 19$, the critical values are 6.84 and 38.58.

(b) $\alpha = 0.05$, $n = 12$: $\alpha/2 = 0.025$, $1 - \alpha/2 = 0.975$; for $n - 1 = 11$, the critical values are 3.82 and 21.92.

(c) $\alpha = 0.10$, $n = 15$: $\alpha/2 = 0.05$, $1 - \alpha/2 = 0.95$; for $n - 1 = 14$, the critical values are 6.57 and 23.68.

Example 3.3 (Critical values, upper one-sided variance test)**.** Consider the test of $H_0 : \sigma^2 = 10$ against $H_1 : \sigma^2 > 10$. What are the critical values for the test statistic χ_0^2 for the following significance levels and sample sizes?

(a) $\alpha = 0.01$ and $n = 20$;

(b) $\alpha = 0.05$ and $n = 12$;

(c) $\alpha = 0.10$ and $n = 15$.

Solution.

(a) $\alpha = 0.01$, $n - 1 = 19$: critical value $\chi^2_{\alpha,n-1} = \chi^2_{0.01,19} = 36.19$.

(b) $\alpha = 0.05$, $n - 1 = 11$: critical value $\chi^2_{0.05,11} = 19.68$.

(c) $\alpha = 0.10$, $n - 1 = 14$: critical value $\chi^2_{0.1,14} = 21.06$.

Example 3.4 (Critical values, lower one-sided variance test)**.** For $H_0 : \sigma^2 = 5$ against $H_1 : \sigma^2 < 5$, find the critical values.

(a) $\alpha = 0.01$, $n - 1 = 19$: use $1 - \alpha = 0.99$; $\chi^2_{0.99,19} = 7.63$.

(b) $\alpha = 0.05$, $n - 1 = 11$: use $1 - \alpha = 0.95$; $\chi^2_{0.95,11} = 4.57$.

(c) $\alpha = 0.10$, $n - 1 = 14$: use $1 - \alpha = 0.90$; $\chi^2_{0.90,14} = 7.79$.

Example 3.5 (Ice-hockey performance after electrostimulation)**.** The data from *Medicine and Science in Sports and Exercise* considered ice hockey player performance after electrostimulation training. In summary there were 17 players and the sample standard deviation of performance was 0.09 seconds.

(a) Is there strong evidence to conclude that the standard deviation of performance time exceeds the historical value of 0.75 seconds? Use $\alpha = 0.05$. Find the P-value for this test.

(b) Discuss how part (a) could be answered by constructing a 95% one-sided confidence interval for σ.

Solution. $n = 17$, $s = 0.09$ sec, $H_0 : \sigma^2 = 0.75$, $H_1 : \sigma > 0.75$, $\alpha = 0.05$.

(a)

$$\chi_0^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{16 \times (0.09)^2}{(0.75)^2} = 0.2304.$$

$$\chi^2_{\text{critical}} = \chi^2_{0.05,\,n-1} = \chi^2_{0.05,16} = 26.30.$$

Since $\chi_0^2 < \chi^2_{\text{critical}}$ (i.e. $0.2304 < 26.30$), do not reject H_0 (fail to reject H_0). Hence σ^2 may be 0.5625, or $\sigma = 0.75$.

P-value: the probability above χ_0^2 here is > 0.995 (based on the calculated chi-square value; check the table for which α value is nearest, and the P-value for $H_1 : \sigma > \sigma_0$ will be greater than that α value).

(b) A $100(1-\alpha)\%$ (here 95%) lower confidence bound for σ^2 is

$$\frac{(n-1)s^2}{\chi^2_{\alpha,\,n-1}} \le \sigma^2,$$

with $\alpha = 0.05$, $n = 16$ (as used in the notes), $\chi^2_{0.05,16} = 26.30$ (and $\chi^2_{0.95,16} = 7.96$ for reference):

$$\frac{(17-1)(0.09)^2}{26.30} \le \sigma^2 \;\Rightarrow\; \boxed{0.004 \le \sigma^2} \quad \text{or} \quad \boxed{0.07 \le \sigma.}$$

Example 3.6 (Two-sided variance test — concepts)**.** (i) What are the hypothesis statements and the statistic to be used in a two-sided test of the variance of a population? What are the assumptions to use this statistic, and what is the rejection region for this test?

(ii) What is the value x such that a chi-square random variable with 15 degrees of freedom takes on at most a value of x with 95% likelihood?

Solution.

(i) Hypotheses (two-sided, with one-sided variants listed):

$$H_0 : \sigma^2 = \sigma_0^2 \quad \text{vs.} \quad H_1 : \sigma^2 \ne \sigma_0^2 \qquad \big(H_1 : \sigma^2 < \sigma_0^2 \;\text{ or }\; H_1 : \sigma^2 > \sigma_0^2\big).$$

Statistic: $\chi_0^2 = \dfrac{(n-1)s^2}{\sigma^2}$. Rejection region: χ_0^2 outside $\big(\chi^2_{1-\alpha/2,\,n-1},\ \chi^2_{\alpha/2,\,n-1}\big)$, i.e. $\chi_0^2 > \chi^2_{\alpha,n-1}$ or $\chi_0^2 < \chi^2_{1-\alpha,n-1}$ (as appropriate). *Assumption:* the population is normally distributed.

(ii)

$$\mathrm{P}(\chi^2_{15} < x) = 0.95 \;\Rightarrow\; \mathrm{P}(\chi^2_{15} > x) = 0.05 \;\Rightarrow\; x = \chi^2_{0.05,\,15}.$$

$$\boxed{x = 25.}$$

Example 3.7 (Variance greater than 10 — sample data)**.** Test at 10% significance whether the variance of the population is greater than 10, with the help of the sample data

$$24,\ 22,\ 25,\ 23,\ 20,\ 23,\ 21,\ 22,\ 25,\ 22.$$

How can you answer this question with the help of confidence intervals?

Solution. First compute the sample statistics ($n = 10$):

$$\bar{X} = 22.7, \qquad s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = 2.6778, \qquad s = 1.63639.$$

Hypotheses: $H_0 : \sigma^2 = 10$ vs. $H_1 : \sigma^2 > 10$, $\alpha = 0.1$.

Test statistic:

$$\chi_0^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{9 \times (1.63639)^2}{10} = 2.4.$$

Critical value: $\chi^2_{0.1,9} = 14.68$ (from tables).

Since $\chi_0^2 < \chi^2_{0.1,9}$ (i.e. $2.4 < 14.68$), do not reject H_0; conclude that σ^2 may be $= 10$.

(b) 90% lower limit of σ^2: a $100(1-\alpha)\%$ bound gives

$$\sigma^2 \ge \frac{(n-1)s^2}{\chi^2_{\alpha,\,n-1}} = \frac{9 \times (1.6364)^2}{14.68} = 1.642.$$

The interval suggests that σ^2 may be 10.

Example 3.8 (Tyre life — standard deviation)**.** A research engineer for a tyre manufacturer is investigating tyre life for a new rubber compound and has built 16 tyres and tested them to end-of-life in a road test. The sample mean and standard deviation are 60139.7 and 3645.94 km.

(i) Can you conclude, using $\alpha = 0.05$, that the standard deviation of tyre life exceeds 200 km? State any necessary assumption about the underlying distribution of the data.

(ii) Answer question (i) by constructing an appropriate confidence interval.

Solution. Given $\bar{X} = 60139.7$ km, $s = 3645.94$ km, $n = 16$, $\alpha = 0.05$, $H_0 : \sigma = 200$, $H_1 : \sigma > 200$.

(i)

$$\chi_0^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{15 \times (3645.94)^2}{200^2} = 4984.8 \approx 4985.$$

Critical value: $\chi^2_{0.05,15} = 25$. Since $\chi_0^2 > \chi^2_{0.05,15}$, we reject H_0; therefore $\sigma > 200$. *Assumption:* the tyre life is normally distributed.

(ii) A $100(1-\alpha)\%$ (here 95%) lower-bound confidence interval on σ is

$$\sqrt{\frac{(n-1)s^2}{\chi^2_{\alpha,\,n-1}}} \leq \sigma \;\Rightarrow\; \sigma \geq \sqrt{\frac{15 \times (3645.94)^2}{25}} \;\Rightarrow\; \sigma \geq 2824.13.$$

Hence we conclude that σ is far greater than 200.

Chapter 4

Tests on the Mean of a Normal Distribution, Variance Unknown

In practice the population variance σ^2 is rarely known. When it must be estimated from the sample standard deviation s, the appropriate reference distribution is Student's t rather than the standard normal. This chapter develops the t-test for the mean.

4.1 The t Distribution

If $X_1, X_2, \ldots, X_n$ is a random sample from a normal distribution with mean μ and variance σ^2, the random variable

$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

has a t distribution with $n - 1$ degrees of freedom.

The general appearance of the t distribution is similar to the standard normal distribution in that both distributions are symmetric and unimodal, and the maximum ordinate value is reached when the mean $\mu = 0$. However, the t distribution has heavier tails than the normal; that is, it has more probability in the tails than the normal distribution.

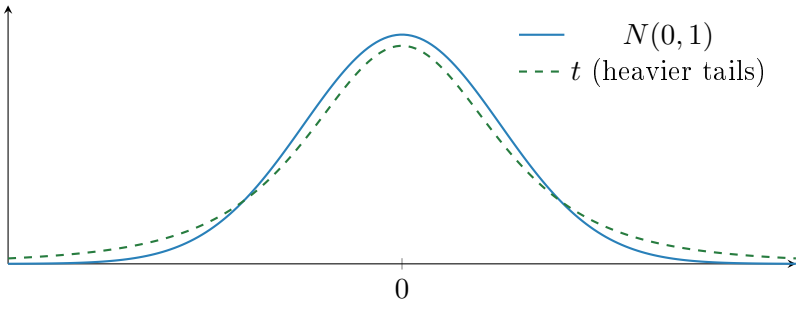


Figure 4.1: The t distribution compared to the standard normal: same symmetric, unimodal shape but heavier tails.

4.2 Hypothesis Tests on the Mean (σ unknown)

The hypotheses are $H_0 : \mu = \mu_0$, $H_1 : \mu \neq \mu_0$.

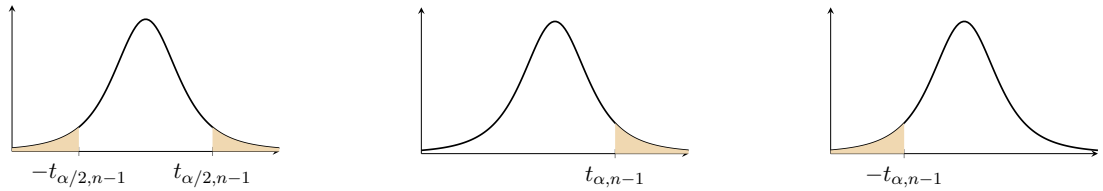
Test Statistic (σ unknown)

$$T_0 = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}.$$

If the null hypothesis is true, T_0 has a t distribution with $n-1$ degrees of freedom. We use the t percentage points $-t_{\alpha/2,n-1}$ and $t_{\alpha/2,n-1}$ as the boundaries of the critical region.

Table 4.1: Rejection criteria for the t-test on the mean.

Alternative hypothesis	**Rejection criterion**
$H_1 : \mu \neq \mu_0$	$t_0 > t_{\alpha/2,\, n-1}$ or $t_0 < -t_{\alpha/2,\, n-1}$
$H_1 : \mu > \mu_0$	$t_0 > t_{\alpha,\, n-1}$
$H_1 : \mu < \mu_0$	$t_0 < -t_{\alpha,\, n-1}$



(a) $H_1 : \mu \neq \mu_0$ (b) $H_1 : \mu > \mu_0$ (c) $H_1 : \mu < \mu_0$

Figure 4.2: The reference t_{n-1} distribution with critical regions (shaded) for the three alternative hypotheses.

4.3 P-value for a t-test

The P-value for a t-test is the smallest level of significance at which the null hypothesis would be rejected. It is the tail area beyond the value of the test statistic for a one-sided test, or twice this area for a two-sided test.

Computation of the exact P-value directly from a t-table is usually impossible. However, it is easy to find upper and lower bounds on the P-value from the table.

Example 4.1 (Bounding a P-value from the t-table)**.** Consider the following critical values for a t-test based on 14 degrees of freedom.

Critical value	0.258	0.692	1.345	1.761	2.145	2.624	2.977
Tail area	0.4	0.25	0.1	0.05	0.025	0.01	0.005

If in a problem $t_0 = 2.72$, this lies between the values 2.624 and 2.977; therefore the P-value must be between 0.01 and 0.005. These are effectively the upper and lower bounds on the P-value.

4.4 Worked Problems

Example 4.2 (Critical values for a two-sided t-test)**.** A hypothesis will be used to test that a population mean equals 7 against the alternative that the population mean does not equal 7

with unknown variance σ. What are the critical values for the test statistic T_0 for the following significance levels and sample sizes?

(a) $\alpha = 0.01$ and $n = 20$;

(b) $\alpha = 0.05$ and $n = 12$;

(c) $\alpha = 0.10$ and $n = 15$.

Solution. $H_0 : \mu = 7$, $H_1 : \mu \neq 7$.

(a) $\alpha = 0.01$, $\alpha/2 = 0.005$, $n - 1 = 19$: $t_{\alpha/2,n-1} = t_{0.005,19} = 2.861$; $-t_{0.005,19} = -2.861$; so $-2.861 \leq t_0 \leq 2.861$.

(b) $\alpha = 0.05$, $n = 12$, $\alpha/2 = 0.025$: $t_{\alpha/2,n-1} = 2.201$; so $-2.201 \leq t_0 \leq 2.201$.

(c) $\alpha = 0.10$, $n = 15$, $\alpha/2 = 0.05$, $n - 1 = 14$: $t_{0.05,14} = 1.761$; so $-1.761 \leq t_0 \leq 1.761$.

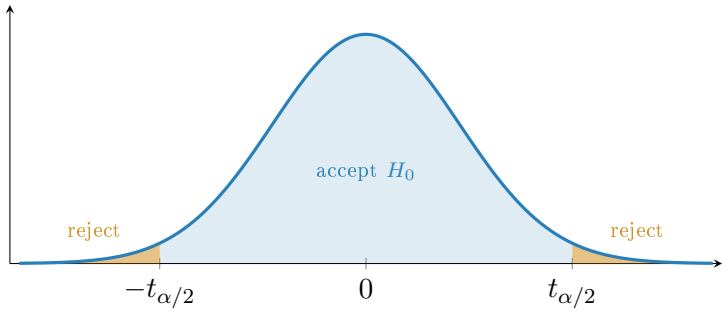


Figure 4.3: Two-sided rejection region for the t-test: reject H_0 when $|t_0| > t_{\alpha/2,n-1}$.

Example 4.3 (Critical values for an upper one-sided t-test)**.** $H_0 : \mu = 10$, $H_1 : \mu > 10$, variance σ unknown. Find the critical value.

(a) $\alpha = 0.01$, $n = 20$;

(b) $\alpha = 0.05$, $n = 12$;

(c) $\alpha = 0.10$, $n = 15$.

Solution.

(a) $\alpha = 0.01$, $n = 20$: critical value $t_{\alpha,n-1} = t_{0.01,19} = 2.539$; so $t_0 > 2.539$.

(b) $\alpha = 0.05$, $n = 12$: $t_{0.05,11} = 1.796$; so $t_0 > 1.796$.

(c) $\alpha = 0.10$, $n = 15$: $t_{0.1,14} = 1.345$; so critical value > 1.345.

Example 4.4 (Critical values for a two-sided t-test at $\mu = 5$)**.** $H_0 : \mu = 5$, $H_1 : \mu \neq 5$, variance σ unknown. (a) $\alpha = 0.01$, $n = 20$; (b) $\alpha = 0.05$, $n = 12$; (c) $\alpha = 0.10$, $n = 15$.

Solution (answers).

1. Critical value < 2.539 (i.e. $|t_0|$ compared to 2.539).
2. Critical value < 1.796.
3. Critical value < 1.345.

Example 4.5 (Bounding P-values, two-sided t-test)**.** For the hypothesis test $H_0 : \mu = 7$ against $H_1 : \mu \neq 7$ with variance unknown and $n = 20$, approximate the P-value for each of the following test statistics: (a) $t_0 = 2.05$, (b) $t_0 = -1.84$, (c) $t_0 = 0.4$.

Solution. Use $n - 1 = 19$ degrees of freedom.

(a) $t_0 = 2.05$: for $n - 1 = 19$, from the tables $t_0 = 2.05$ lies between 1.729 and 2.093, i.e. for $\alpha/2$ between 0.05 and 0.025 (so α between 0.1 and 0.05). Hence $0.05 \leq P \leq 0.1$.

(b) $t_0 = -1.84$: also lies with $\alpha/2$ between 0.05 and 0.025; hence $0.05 \leq P \leq 0.1$.

(c) $t_0 = 0.4$: with $\alpha/2$ between 0.40 and 0.25; hence $0.5 \leq P \leq 0.8$.

Example 4.6 (Bounding P-values, lower one-sided t-test)**.** For the hypothesis test $H_0 : \mu = 5$ against $H_1 : \mu < 5$ with variance unknown and $n = 12$, approximate the P-value for each of the following test statistics: (a) $t_0 = 2.05$, (b) $t_0 = -1.84$, (c) $t_0 = 0.4$.

Solution. Use $n - 1 = 11$ degrees of freedom.

(a) $t_0 = 2.05$: lies with tail area between 0.05 and 0.025, so $0.95 \leq P \leq 0.975$.

(b) $t_0 = -1.84$: lies with tail area between 0.05 and 0.025, so $0.025 \leq P \leq 0.05$.

(c) $t_0 = 0.4$: tail area between 0.40 and 0.25; with $1 - \alpha$ giving 0.6 to 0.75, so $0.6 \leq P \leq 0.75$.

Example 4.7 (Machine installation time)**.** The time in hours to install and commission a machine is as follows for a sample of 10 machines chosen at random:

$$192,\ 201,\ 190,\ 183,\ 185,\ 176,\ 191,\ 214,\ 220,\ 205.$$

(i) If the above data is assumed to come from a normal population, can you conclude that the population mean is greater than 205 hours at a 5% significance level?

(ii) Derive an interval estimate to answer the same question.

Solution. $H_0 : \mu = 205$, $H_1 : \mu > 205$ hr, $\alpha = 0.05$. Test statistic $t_0 = \dfrac{\bar{x} - \mu}{s/\sqrt{n}}$.

Sample statistics:

$$\bar{X} = 195.7, \qquad s^2 = \frac{1772.1}{10 - 1} = 196, \qquad s = 14.032.$$

$$t_0 = \frac{195.7 - 205}{14.0321/\sqrt{10}} = -2.0959, \qquad t_{\alpha,n-1} = t_{0.05,9} = 1.833.$$

Since $t_0 < t_{\alpha,n-1}$, do not reject H_0: μ may be 205, not > 205 hours.

(ii) Interval estimate. Two-sided:

$$\bar{X} \pm t_{\alpha/2}\,\frac{s}{\sqrt{n}} = 195.7 \pm 2.262 \cdot \frac{14.0321}{\sqrt{10}} = (185.66,\ 205.74).$$

Lower bound:

$$\mu \geq \bar{X} - t_{\alpha}\,\frac{s}{\sqrt{n}} = 195.7 - 1.833 \cdot \frac{14.052}{\sqrt{10}} = 187.57.$$

Both the two-sided and the lower-bound confidence intervals include 205 hr; therefore we cannot reject the claim that $\mu = 205$ hr.

Example 4.8 (Overweight documentary)**.** A television documentary on overrating weight claimed that Americans are about 10 pounds overweight on average. To test the claim, 18 randomly selected individuals were examined; their average excess weight was found to be 12.4 pounds and the sample standard deviation was 2.7 pounds. At a significance level of 0.01, is there any reason to doubt the validity of the claimed 10-pound value?

Solution. $\bar{X} = 12.4$, $s = 2.7$, $\alpha = 0.01$, $n = 18$, $H_0 : \mu = 10$ pounds vs. $H_1 : \mu > 10$ lb.

$$t_0 = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{12.4 - 10}{2.7/\sqrt{18}} = 3.77, \qquad t_{\alpha,n-1} = t_{0.01,17} = 2.567.$$

Since $t_0 > t_{\alpha,n-1}$, reject H_0. The data show that Americans are overweight by more than 10 pounds on average, and the claim of just 10 lb overweight is not true.

Example 4.9 (Probability from the t distribution)**.** State the probability of a t-random variable with 25 degrees of freedom taking on any value between 2.06 and 3.45.

Solution.

$$\mathrm{P}(2.06 < T_{25} < 3.45) = \mathrm{P}(T > 2.06) - \mathrm{P}(T > 3.45) = 0.025 - 0.001 = 0.024.$$

Chapter 5

Testing for Goodness of Fit

The tests of the previous chapters assumed a known form of distribution (normal) and tested its parameters. We now consider a different kind of hypothesis: we do *not* know the underlying distribution of the population, and we wish to test the hypothesis that a particular distribution will be satisfactory as a population model.

5.1 The Chi-Square Goodness-of-Fit Test

Hypothesis-testing procedures can be of two types: problems in which the probability distribution is known and the hypotheses involve the parameters of the distribution; and another kind, in which we do not know the underlying distribution of the population and we wish to test the hypothesis that a particular distribution will be satisfactory as a population model.

The goodness-of-fit test procedure requires a random sample of size n from the population whose probability distribution is unknown. These n observations are arranged in a frequency histogram having k bins or class intervals. Let O_i be the observed frequency in the ith class interval. From the hypothesized probability distribution, we compute the expected frequency in the ith class interval, denoted E_i.

Goodness-of-Fit Test Statistic

$$\chi_0^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}.$$

It can be shown that, if the population follows the hypothesized distribution, χ_0^2 has approximately a chi-square distribution with $k - p - 1$ degrees of freedom, where p represents the number of parameters of the hypothesized distribution estimated by sample statistics. This approximation improves as n increases.

Rejection Criterion

We reject the hypothesis that the distribution of the population is the hypothesized distribution if the calculated value of the test statistic

$$\chi_0^2 > \chi_{\alpha,\, k-p-1}^2.$$

On small expected frequencies

If the expected frequencies are too small, the test statistic χ_0^2 will not reflect the departure of observed from expected, but only the small magnitude of the expected frequencies. There is no general agreement regarding the minimum value of expected frequencies, but values of 3, 4, and 5 are widely used as minimal. If an expected frequency is too small, it can be combined with the expected frequency in an adjacent class interval. The corresponding observed frequencies would then also be combined, and k would be reduced by 1. Class intervals are not required to be of equal width.

5.2 Worked Problems

Example 5.1 (Defects in printed circuit boards — Poisson fit)**.** The number of defects in printed circuit boards is hypothesized to follow a Poisson distribution. A random sample of $n = 60$ printed circuit boards has been collected, and the following number of defects observed. Perform a test of hypothesis.

Table 5.1: Observed defect frequencies.

Number of defects	Observed frequency
0	32
1	15
2	9
3	4

Solution (eight-step procedure).

1. The variable of interest is the form of the distribution of defects in printed circuit boards.
2. H_0: The form of the distribution of defects is Poisson.
3. H_1: The form of the distribution of defects is not Poisson.
4. $\alpha = 0.05$.
5. Test statistic χ_0^2. The mean of the distribution is not known, so to estimate it we use the sample average:

$$\hat{\lambda} = \frac{32 \cdot 0 + 1 \cdot 15 + 2 \cdot 9 + 3 \cdot 4}{60} = 0.75.$$
$$\therefore \ \lambda = 0.75.$$

Computations. We compute p_i, the theoretical (hypothesized) probability associated with the ith class interval. Since each class interval corresponds to a number of defects, we find p_i as follows:

$$p_1 = \mathrm{P}(X = 0) = \frac{e^{-\lambda}\lambda^0}{0!} = \frac{e^{-0.75}\,0.75^0}{0!} = 0.472,$$
$$p_2 = \mathrm{P}(X = 1) = \frac{e^{-0.75}\,0.75^1}{1!} = 0.354,$$
$$p_3 = \mathrm{P}(X = 2) = \frac{e^{-0.75}\,0.75^2}{2!} = 0.133,$$
$$p_4 = \mathrm{P}(X \geq 3) = 1 - (p_1 + p_2 + p_3) = 0.041.$$

The expected frequencies are computed by multiplying the sample size $n = 60$ by each probability, i.e. $E_i = n \cdot p_i$.

Table 5.2: Goodness-of-fit computation for the Poisson model (classes 2 and 3^+ pooled).

No. of defects	**Prob.**	E_i	O_i	$(O_i - E_i)^2/E_i$
0	0.472	28.32	32	0.4781
1	0.354	21.24	15	1.833
2	0.133	7.8 } 10.44	9 } 13	0.6277
3 (or more)	0.041	2.46 }	4 }	
			$\chi^2_{\text{cal}} =$	2.94

6. Since the mean is estimated, $p = 1$; after pooling there are $k = 3$ class intervals, so the degrees of freedom are $k - p - 1 = 3 - 1 - 1 = 1$:

$$\chi^2_{\text{tab}} = \chi^2_{0.05,\,k-p-1} = \chi^2_{0.05,1} = 3.84.$$

8. **Conclusion:** Since $\chi^2_0 = 2.94 < \chi^2_{0.05,1} = 3.84$, we fail to reject the null hypothesis that the distribution of defects in printed circuit boards is Poisson.

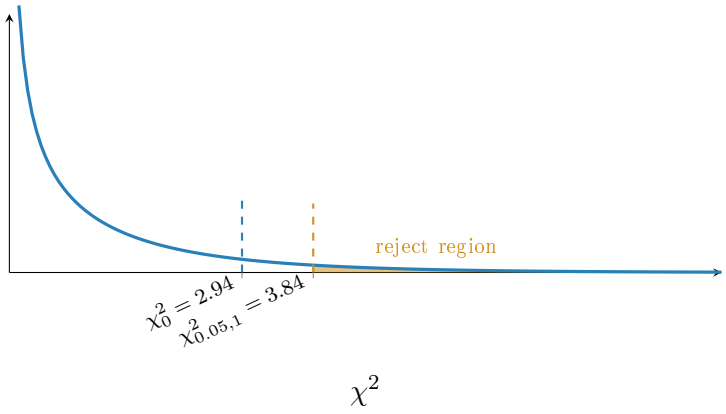


Figure 5.1: Chi-square distribution with 1 degree of freedom. The statistic $\chi^2_0 = 2.94$ falls left of the critical value 3.84, so H_0 is not rejected.

Example 5.2 (Safety incidents — Poisson fit)**.** X, the number of reported safety incidents per day on a production shop floor, is recorded for 90 days as follows.

Table 5.3: Observed safety-incident frequencies over 90 days.

x_i: no. of accidents	0	1	2	3	≥ 4
No. of days with x_i accidents	54	23	9	3	1

Does the data follow a Poisson distribution? Test at 5% significance level.

Solution.

1. Point of interest: the data of accidents and their distribution.

2. H_0: Data follow a Poisson distribution.
3. H_1: Data do not follow a Poisson distribution.
4. $\alpha = 0.05$.
5. Test statistic $\chi_0^2 = \sum \dfrac{(O_i - E_i)^2}{E_i}$.
6. Computations. The estimated mean is

$$\lambda = \frac{0 \cdot 54 + 1 \cdot 23 + 2 \cdot 9 + 3 \cdot 3 + 4 \cdot 1}{90} = 0.6.$$

The Poisson probabilities are

$$p_0 = \mathrm{P}(X = 0) = \frac{e^{-\lambda}\lambda^0}{0!} = \frac{e^{-0.6}\,0.6^0}{0!} = 0.5488,$$

$$p_1 = \mathrm{P}(X = 1) = 0.3293, \qquad p_2 = \mathrm{P}(X = 2) = 0.0988, \qquad p_3 = \mathrm{P}(X = 3) = 0.0198,$$

$$p_4 = \mathrm{P}(X \geq 4) = 1 - 0.5488 - 0.3293 - 0.0988 - 0.0198 = 0.0033.$$

Table 5.4: Goodness-of-fit computation for the safety-incident data (classes 2, 3, ≥ 4 pooled).

x_i	O_i		p_i	E_i		$(O_i - E_i)^2/E_i$
0	54		0.5488	49.39		0.4303
1	23		0.3293	29.64		1.4875
2	9	} 13	0.0988	8.89	} 10.97	0.3757
3	3		0.0198	1.79		
≥ 4	1		0.0033	0.3		
					$\chi_0^2 =$	2.2935

$$\chi^2_{\text{tab}} = \chi^2_{3-1-1,\,0.05} = \chi^2_{0.05,1} = 3.84.$$

7. Rejection criterion: since $\chi_0^2 < \chi^2_{\text{tab}}$ ($2.2935 < 3.84$), the null hypothesis is *not rejected* at the 5% level. The data are consistent with a Poisson distribution.

Example 5.3 (Radar components — binomial fit)**.** The production of a radar component is checked by a sample of four. The following table shows the number of defectives found in 300 samples.

Table 5.5: Defectives in 300 samples of size four.

No. of defectives	0	1	2	3	4
No. of samples	90	130	58	20	2

Fit the binomial distribution for the data and test the goodness of fit at a level of significance of 0.05.

Solution. The mean of the observed distribution is

$$\mu = np = \frac{0 \cdot 90 + 1 \cdot 130 + 2 \cdot 58 + 3 \cdot 20 + 4 \cdot 2}{300} = \frac{314}{300} = 1.047.$$

Since $np = 1.047 = 4p$, we estimate $p = 1.047/4 = 0.262$. The binomial probabilities are

$$\mathrm{P}(X = x) = \binom{n}{x} p^x (1-p)^{n-x} = \binom{4}{x} (0.262)^x (0.738)^{4-x},$$

so that, for example, $\mathrm{P}(X = 0) = \binom{4}{0}(0.262)^0(0.738)^4 = 0.2966$.

Table 5.6: Goodness-of-fit computation for the binomial model (classes 3 and 4 pooled).

Value (X)	Prob.	Observed freq.	Expected	$(O_i - E_i)^2/E_i$
0	0.2966	90	88.98	0.0117
1	0.4212	130	126.36	0.1049
2	0.2243	58	67.29	1.2826
3	0.0529	20 } 22	15.87 } 17.28	1.2867
4	0.0047	2 }	1.41 }	
			$\sum \chi^2 =$	2.6859

The tabulated value (one parameter p estimated, $k = 5$ classes; note the notes use $\chi^2_{0.05,4-1-1} = \chi^2_{0.05,2}$):

$$\chi^2_{\text{tab}} = \chi^2_{0.05,\,n-k-1} = \chi^2_{0.05,\,4-1-1} = \chi^2_{0.05,2} = 5.99.$$

Since $\chi^2_{\text{cal}} = 2.6859 < \chi^2_{\text{tab}} = 5.99$, we fail to reject H_0. We can conclude the distribution is binomial.

Example 5.4 (Octahedral die — test for fairness (uniform fit))**.** Nadir is testing an octahedral die to see if it is biased. The results are given as

Table 5.7: Observed scores of an octahedral die.

Score	1	2	3	4	5	6	7	8
Frequency	7	10	11	9	12	10	14	7

Test the hypothesis that the die is fair.

Solution. H_0: Die is fair; H_1: Die is not fair. Under H_0 the distribution is uniform (all die sides have equal probability 1/8). $\alpha = 0.05$. Degrees of freedom $\nu = n - k - 1$; no parameter is estimated, so $k = 0$ and $\nu = 8 - 1 = 7$.

$$\chi^2_{\text{tab}} = \chi^2_{0.05,7} = 14.067.$$

The expected value in each class is

$$E_i = \frac{1}{8}(7 + 10 + 11 + 9 + 12 + 10 + 14 + 7) = 10.$$

Table 5.8: Goodness-of-fit computation for the fair-die (uniform) model.

Score	Frequency	E_i	$(O_i - E_i)^2/E_i$
1	7	10	0.9
2	10	10	0
3	11	10	0.1
4	9	10	0.1
5	12	10	0.4
6	10	10	0
7	14	10	1.6
8	7	10	0.9
		$\chi^2_{\text{cal}} =$	4.0

Since $\chi^2_{\text{cal}} < \chi^2_{\text{tab}}$, we fail to reject H_0, i.e. the die is fair. There is no sufficient evidence at the 5% significance level to suggest that the die is not fair.

Example 5.5 (Galvanized steel coils — Poisson fit)**.** Let X denote the number of flaws observed on a large coil of galvanized steel. Seventy-five coils are inspected, and the following data were observed for the values of X.

Table 5.9: Observed flaw counts on 75 coils.

Values (X)	1	2	3	4	5	6	7	8
Observed frequency	1	11	8	13	11	12	10	9

(a) Does the assumption of a Poisson distribution seem appropriate as a probability model for this data? Use $\alpha = 0.01$.

(b) Calculate the P-value for this test.

Solution. H_0: Poisson distribution is appropriate for the data; H_1: Data do not follow a Poisson distribution; $\alpha = 0.01$. The estimated mean is

$$\lambda = \frac{1 \cdot 1 + 2 \cdot 11 + 3 \cdot 8 + 4 \cdot 13 + 5 \cdot 11 + 6 \cdot 12 + 7 \cdot 10 + 8 \cdot 9}{75} = 4.9067.$$

Using $\mathrm{P}(X = x) = e^{-\lambda}\lambda^x/x!$, the expected frequencies $E_i = 75\,p_i$ are computed as below (the first two classes and the last two are pooled so that expected counts are not too small).

Table 5.10: Goodness-of-fit computation for the Poisson model (coils); classes 1–2 and 7–8 pooled.

Values	O_i		E_i		$(O_i - E_i)^2/E_i$
1	1	12	2.7214	9.3987	0.7200
2	11		6.6770		
3	8		10.9213		0.7822
4	13		13.3977		0.0118
5	11		13.1485		0.3511
6	12		10.7533		0.1445
7	10	19	7.5381	12.618	3.8449
8	9		4.6237		
				$\chi^2_{\text{cal}} =$	5.8545

After pooling there are 6 classes and one parameter (λ) estimated, so the degrees of freedom are $6 - 1 - 1 = 4$:

$$\chi^2_{\text{tab}} = \chi^2_{0.01,\,6-1-1} = \chi^2_{0.01,4} = 13.28.$$

Since $\chi^2_{\text{cal}} < \chi^2_{\text{tab}}$ ($5.8545 < 13.28$), we fail to reject the null hypothesis, i.e. the data follow a Poisson distribution.

(b) The P-value for this test is $0.1 < P\text{-value} < 0.50$.

Chapter 6

Analysis of Statistical Intervals: Single Sample

Hypothesis testing decides between two statements about a parameter. A companion approach is *interval estimation*: rather than a single point estimate, we report a range of plausible values together with a confidence level. This chapter develops confidence intervals for the mean (variance known and unknown) and for the variance/standard deviation of a normal population, and shows how they answer the same questions as the tests of earlier chapters.

6.1 Confidence Intervals: Basic Ideas

Confidence Interval

An interval estimate for a population parameter is called a **confidence interval**. We cannot be certain that the interval contains the true, unknown population parameter — we only use a sample from the full population to compute the point estimate and the interval. However, the confidence interval is constructed so that we have confidence that it does contain the unknown population parameter. Confidence intervals are widely used in engineering and the sciences.

Interval estimates can be of three types:

1. A *confidence interval* bounds population or distribution parameters (such as a mean viscosity).
2. A *tolerance interval* bounds a selected proportion of a distribution.
3. A *prediction interval* bounds future observations from the population or distribution.

6.2 Confidence Interval on the Mean, Variance Known

Suppose that $X_1, X_2, \ldots, X_n$ is a random sample from a normal distribution with unknown mean μ and known variance σ^2. The sample mean $\bar{X}$ is normally distributed with mean μ and variance σ^2/n. Standardising,

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

has a standard normal distribution.

A confidence interval estimate for μ is an interval of the form $l \le \mu \le u$, where the endpoints l and u are computed from the sample data. Because different samples produce different values of l and u, these endpoints are values of random variables L and U respectively. Suppose that we can determine values of L and U such that the following probability statement is true:

$$\mathrm{P}\{L \le \mu \le U\} = 1 - \alpha, \qquad 0 \le \alpha \le 1.$$

There is a probability of $1-\alpha$ of selecting a sample for which the confidence interval will contain the true value of μ. Once we have selected the sample (so that $X_1 = x_1, \ldots, X_n = x_n$) and computed l and u, the resulting confidence interval for μ is $l \le \mu \le u$.

Confidence Limits and Coefficient

The endpoints (or bounds) l and u are called the **lower** and **upper confidence limits** respectively, and $1 - \alpha$ is called the **confidence coefficient**.

Because $Z = (\bar{X} - \mu)/(\sigma/\sqrt{n})$ has a standard normal distribution, we may write

$$\mathrm{P}\left\{-Z_{\alpha/2} \le \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \le Z_{\alpha/2}\right\} = 1 - \alpha.$$

Manipulating the terms inside the brackets — (1) multiplying through by $\sigma/\sqrt{n}$, (2) subtracting $\bar{X}$ from each term, and (3) multiplying through by -1 — we obtain

$$\mathrm{P}\left\{\bar{X} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le \mu \le \bar{X} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right\} = 1 - \alpha.$$

CI on the Mean, Variance Known

If $\bar{x}$ is the sample mean of a random sample of size n from a normal population with known variance σ^2, a $100(1-\alpha)\%$ confidence interval on μ is

$$\bar{x} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le \mu \le \bar{x} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}},$$

where $Z_{\alpha/2}$ is the upper $100\alpha/2\%$ point of the standard normal distribution.

Note

For a fixed sample size n and standard deviation σ, the higher the confidence level, the longer the resulting confidence interval. The length of the confidence interval is a measure of the precision of estimation.

Choice of sample size

Sample Size for a Specified Error E

If $\bar{x}$ is used as an estimate of μ, we can be $100(1-\alpha)\%$ confident that the error $|\bar{x} - \mu|$ will not exceed a specified amount E when the sample size is

$$n = \left(\frac{Z_{\alpha/2}\,\sigma}{E}\right)^2.$$

If the right-hand side of this equation is not an integer, it must be rounded up. This will ensure that the level of confidence does not fall below $100(1-\alpha)\%$. Notice that $2E$ is the length of the resulting confidence interval.

One-sided confidence bounds

It is possible to obtain one-sided confidence bounds for μ by setting either $l = -\infty$ or $u = \infty$ and replacing $Z_{\alpha/2}$ by Z_α.

One-Sided Confidence Bounds (σ known)

A $100(1-\alpha)\%$ **upper** confidence bound for μ is

$$\mu \leq u = \bar{x} + Z_\alpha \frac{\sigma}{\sqrt{n}},$$

and a $100(1-\alpha)\%$ **lower** confidence bound for μ is

$$\bar{x} - Z_\alpha \frac{\sigma}{\sqrt{n}} = l \leq \mu.$$

6.3 Worked Problems: Mean, Variance Known

Example 6.1 (CI and sample size for a small sample)**.** For the sample $(5, 2, 3, 1, 0, 0, 1, 3, 3)$ from a population:

(i) Construct a 90% confidence interval on the population mean if the population standard deviation is known to be 2.

(ii) What should be the sample size in order to keep the width of the confidence interval to less than 2?

Solution. $100(1-\alpha)\%$ CI: $\mu \in \bar{x} \pm Z_{0.05}\frac{\sigma}{\sqrt{n}}$. The sample mean is

$$\bar{X} = \frac{5+2+3+1+0+0+1+3+3}{9} = \frac{18}{9} = 2.$$

(i)

$$\text{CI} = 2 \pm \frac{1.645 \cdot 2}{\sqrt{9}} = 2 \pm \frac{1.645 \cdot 2}{3},$$

giving upper $= 3.0967$ and lower $= 0.9033$:

$$\boxed{0.9033 \leq \mu \leq 3.0967.}$$

(ii)

$$\text{width} = \frac{2\, Z_{0.05}\, \sigma}{\sqrt{n}} = \frac{2 \cdot 1.645 \cdot 2}{\sqrt{n}} < 2 \;\Rightarrow\; \sqrt{n} > \frac{2 \cdot 1.645 \cdot 2}{2} = 3.29,$$

$$\Rightarrow\; n \geq 10.824 \;\Rightarrow\; \boxed{n \geq 11.}$$

Example 6.2 (Overfill in beverage bottles)**.** For the sample $(5, 2, 3, 1, 4, 2, 1, 3, 3, 5)$ from a population representing the extent of overfill in sealed beverage bottles in ml:

(a) Construct a 95% confidence upper bound on the population mean if the population variance is known to be 5. What should be the range of sample size in order to keep the width of the confidence interval to less than 1 ml?

(b) If the population variance is unknown, construct a 95% CI on the population variance.

Solution. The sample mean is

$$\bar{X} = \frac{5+2+3+1+4+2+1+3+3+5}{10} = \frac{29}{10} = 2.9.$$

(a) 95% upper bound on the mean. With $Z_{0.05} = 1.645$, $\sigma = \sqrt{5}$:

$$\mu \le \bar{X} + Z_{0.05}\frac{\sigma}{\sqrt{n}} = 2.9 + 1.645 \cdot \frac{\sqrt{5}}{\sqrt{10}} = 2.9 + 1.645\sqrt{\tfrac{5}{10}} \;\Rightarrow\; \boxed{\mu \le 4.063.}$$

This upper bound is the same as the upper limit of the 90% two-sided CI ($\bar{X} + Z_{0.05}\sigma/\sqrt{n}$). For the width requirement,

$$\text{width} = 2 \cdot 1.645\sqrt{\frac{5}{n}} < 1 \;\Rightarrow\; 2 \cdot 1.645\sqrt{5} < \sqrt{n} \;\Rightarrow\; n \ge (2 \cdot 1.645)^2 \cdot 5 = 54.1,$$

so the sample size $n \ge 54$ to keep the CI width below 1 ml.

(b) 95% CI on the variance (population variance unknown), $\alpha = 0.05$, $n = 10$:

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}}.$$

With $s = 1.449$ (so $s^2 = (1.449)^2$), $\chi^2_{0.025,9} = 19.02$ and $\chi^2_{0.975,9} = 2.70$:

$$\frac{9 \cdot (1.449)^2}{19.02} \le \sigma^2 \le \frac{9 \cdot (1.449)^2}{2.70} \;\Rightarrow\; \boxed{0.9937 \le \sigma^2 < 7 \text{ ml}^2,}$$

where s^2 is obtained from $s^2 = \frac{1}{9}\sum(x_i - 2.9)^2$.

Example 6.3 (Standard error and interval estimate)**.** From a population with known standard deviation of 1.65, a sample of 32 items resulted in 34.8 as an estimate of the mean.

(i) Find the standard error of the mean.

(ii) Compute an interval estimate that should include the population mean 99.7 percent of the time.

Solution. (i) Standard error of the mean $= \dfrac{\sigma}{\sqrt{n}} = \dfrac{1.65}{\sqrt{32}} = 0.29.$

(ii) For 99.7% confidence, $Z_{\alpha/2} = Z_{0.0015} = 2.97$:

$$\bar{X} \pm Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = 34.8 \pm 2.97 \cdot \frac{1.65}{\sqrt{32}} \;\Rightarrow\; \boxed{33.93 \le \mu \le 35.67.}$$

Example 6.4 (Cardinal Electricals — light-bulb life)**.** The manager of Cardinal Electricals' light bulb division must estimate the average number of hours that a light bulb made by each bulb machine will last. A sample of 40 light bulbs was selected from a machine, and the average burning time was 1416 hours. The standard deviation of burning time is known to be 30 hours. Construct a 90% CI for the true population mean.

Solution. $n = 40$, $\bar{x} = 1416$, $\sigma = 30$ hr, $\alpha = 0.1$, $Z_{\alpha/2} = 1.645$:

$$\bar{x} \pm Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = 1416 \pm 1.645 \cdot \frac{30}{\sqrt{40}} = (1408.2,\ 1423.8),$$

so $\boxed{1408.2 \le \mu \le 1423.8.}$

Example 6.5 (PC operating-system response time)**.** An operating system for a PC has been studied extensively, and it is known that the standard deviation of the response time following a particular command is $\sigma = 8$ ms. A new version of the operating system is installed, and we wish to estimate the mean response time for the new system, ensuring that a 95% CI for μ has length at most 5 ms.

(a) If response time is normally distributed and $\sigma = 8$ for the new system, what sample size would you recommend?

(b) Suppose the vendor tells us the standard deviation of the new system is smaller, say $\sigma = 6$; give the recommended sample size and comment on the effect of the smaller standard deviation.

Solution. The length of the CI is 5, so length $= 2\,Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$ with CL = 95%, $\alpha = 0.05$, $Z_{\alpha/2} = 1.96$.

(a) $\sigma = 8$:

$$\text{length} = 2 \cdot Z_{0.025} \cdot \frac{8}{\sqrt{n}} = \frac{2 \cdot 1.96 \cdot 8}{\sqrt{n}} = 5 \ \Rightarrow\ \sqrt{n} = \frac{2 \cdot 1.96 \cdot 8}{5} = 6.272,$$

$$n = (6.272)^2 = 39.338 \ \Rightarrow\ \boxed{n = 40.}$$

(b) $\sigma = 6$:

$$\frac{2 \cdot 1.96 \cdot 6}{\sqrt{n}} = 5 \ \Rightarrow\ \sqrt{n} = 4.704 \ \Rightarrow\ n = (4.704)^2 = 22.13 \ \Rightarrow\ \boxed{n = 23.}$$

We observe that as the standard deviation is reduced, the required sample size also reduces.

Example 6.6 (Concrete compressive strength — 95% and 99% CIs)**.** A civil engineer is analysing the compressive strength of concrete. Compressive strength is normally distributed with $\sigma^2 = 1000\,\text{psi}^2$. A random sample of 12 specimens has mean $\bar{x} = 3250$ psi.

(i) Construct a 95% two-sided confidence interval on mean compressive strength.

(ii) Construct a 99% two-sided confidence interval on mean compressive strength.

Solution. $\sigma^2 = 1000$, $n = 12$, $\bar{x} = 3250$.

(i) 95% CI, $\alpha = 0.05$, $\alpha/2 = 0.025$, $Z_{0.025} = 1.96$:

$$\bar{x} \pm 1.96\frac{\sigma}{\sqrt{n}} = 3250 \pm 1.96 \cdot \frac{\sqrt{1000}}{\sqrt{12}} \ \Rightarrow\ \boxed{3232.1 \le \mu \le 3267.89.}$$

(ii) 99% CI, $\alpha = 0.01$, $\alpha/2 = 0.005$, $Z_{\alpha/2} = 2.58$:

$$3250 \pm 2.58 \cdot \frac{\sqrt{1000}}{\sqrt{12}} \ \Rightarrow\ \boxed{3226.4 \le \mu \le 3273.55.}$$

Example 6.7 (Identifying the 95% CI from two intervals)**.** A random sample has been taken from a normal population and two confidence intervals constructed using exactly the same data. The two CIs are (38.02, 61.98) and (39.95, 60.05).

(a) What is the value of the sample mean?

(b) One of these intervals is a 90% CI and the other is a 95% CI. Which one is the 95% CI, and why?

Solution. The CI is given by $\bar{x} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le \mu \le \bar{x} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$, so $\bar{x}$ is the midpoint of the CI.

(a) Using either interval,

$$\bar{X} = \frac{38.02 + 61.98}{2} = \frac{100.00}{2} = 50 \;\Rightarrow\; \boxed{\bar{X} = 50.}$$

(b) For a fixed sample size n and standard deviation σ, the higher the confidence level, the longer (wider) the resulting confidence interval. The two interval widths are

$$(39.95,\ 60.05):\ \text{width} = 20.1 \ \rightarrow\ 90\%\ \text{CI};$$

$$(38.02,\ 61.98):\ \text{width} = 23.96 \ \rightarrow\ 95\%\ \text{CI}.$$

The wider interval (38.02, 61.98) is the 95% CI.

6.4 Confidence Interval on the Mean, Variance Unknown

When the sample is small ($n < 30$ or $n < 40$) and σ^2 is unknown, we must make an assumption about the form of the underlying distribution to obtain a valid CI procedure. A reasonable assumption is that the underlying distribution is normal. Suppose the population of interest has a normal distribution with unknown mean μ and unknown variance σ^2; let $\bar{X}$ and s^2 be the sample mean and variance of a random sample of size n.

To construct a two-sided CI on μ: if the variance were known, we would use $Z = (\bar{X}-\mu)/(\sigma/\sqrt{n})$, which has a standard normal distribution. When σ^2 is unknown, a logical procedure is to replace σ with the sample standard deviation S. The random variable Z now becomes $T = (\bar{X}-\mu)/(S/\sqrt{n})$. (If n is large, one may proceed to use the CI based on the normal distribution.)

The t distribution

Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal distribution with unknown mean μ and unknown variance σ^2. The random variable

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a t distribution with $n-1$ degrees of freedom. The t probability density function is

$$f(x) = \frac{\Gamma[(k+1)/2]}{\sqrt{\pi k}\,\Gamma(k/2)\,[(x^2/k)+1]^{(k+1)/2}}, \qquad -\infty < x < \infty,$$

where k is the number of degrees of freedom. The mean and variance of the t distribution are zero and $k/(k-2)$ for $k > 2$ respectively.

The general appearance of the t distribution is similar to the standard normal: both are symmetric and unimodal, and the maximum ordinate value is reached at the mean $\mu = 0$. However, the t distribution has heavier tails than the normal (more probability in the tails). As the number of degrees of freedom $k \to \infty$, the limiting form of the t distribution is the standard normal. Generally the number of degrees of freedom for t equals the number of degrees of freedom associated with the estimated standard deviation.

Let $t_{\alpha,k}$ be the value of the random variable T with k degrees of freedom above which we find an area (or probability) α; thus $t_{\alpha,k}$ is an upper-tail $100\alpha\%$ point of the t distribution with k degrees of freedom. For example, the t value with 10 degrees of freedom having an area of 0.05 to the right is $t_{0.05,10} = 1.812$, i.e. $\mathrm{P}(T_{10} > t_{0.05,10}) = \mathrm{P}(T_{10} > 1.812) = 0.05$. Since the t distribution is symmetric about zero, $t_{1-\alpha,k} = -t_{\alpha,k}$; e.g. $t_{0.95,10} = -t_{0.05,10} = -1.812$.

CI on the Mean, Variance Unknown

If $\bar{x}$ and s are the mean and standard deviation of a random sample from a normal distribution with unknown variance σ^2, a $100(1-\alpha)\%$ confidence interval on μ is

$$\bar{x} - t_{\alpha/2,\,n-1}\frac{s}{\sqrt{n}} \le \mu \le \bar{x} + t_{\alpha/2,\,n-1}\frac{s}{\sqrt{n}},$$

where $t_{\alpha/2,\,n-1}$ is the upper $100\alpha/2\%$ point of the t distribution with $n-1$ degrees of freedom.

One-sided confidence bounds on the mean of a normal distribution are obtained by using only the appropriate lower or upper confidence limit from the two-sided expression and replacing $t_{\alpha/2,\,n-1}$ by $t_{\alpha,\,n-1}$.

Note

It is not as easy to select a sample size n to obtain a specified length (or precision of estimation) for this CI as it was in the σ-known case, because the length of the interval involves s, n, and $t_{\alpha/2,\,n-1}$. Since the t-percentile depends on the sample size n, an appropriate n can only be obtained through trial and error.

6.5 Worked Problems: Mean, Variance Unknown

Example 6.8 (Reading t-percentiles)**.** Determine the t-percentile that is required to construct each of the following.

(a) Two-sided CI: confidence level = 95%, dof = 12.

(b) Two-sided CI: confidence level = 99.9%, dof = 15.

(c) One-sided CI: confidence level = 99%, dof = 19.

(d) One-sided CI: confidence level = 99.9%, dof = 24 (and dof = 15).

Solution.

(a) $t_{\alpha/2,\,n-1} = t_{0.05/2,\,12} = t_{0.025,12} = 2.179$.

(b) $t_{\alpha/2,\,n-1} = t_{0.001/2,\,15} = 4.073$.

(c) $t_{\alpha,\,n-1} = t_{0.01,19} = 2.539$.

(d) $t_{\alpha,n-1} = t_{0.001,24} = 3.467$; and if dof $= 15$, $t_{0.001,15} = 3.733$.

Example 6.9 (Tyre life — 95% CI on the mean)**.** A research engineer for a tyre manufacturer is investigating tyre life for a new rubber compound and has built 16 tyres and tested them to end-of-life in a road test. The sample mean and standard deviation are 60139.7 and 3645.94 km. Find a 95% CI on mean tyre life.

Solution. $\alpha = 0.05$, $n = 16$, $t_{\alpha/2,n-1} = t_{0.025,15} = 2.131$:

$$\bar{X} - t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} \le \mu \le \bar{X} + t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}$$

$$= 60139.7 - 2.131 \cdot \frac{3645.94}{\sqrt{16}} \le \mu \le 60139.7 + 2.131 \cdot \frac{3645.94}{\sqrt{16}}$$

$$\Rightarrow \boxed{[58197.33,\ 62082.07].}$$

Example 6.10 (Maintenance department — sensor life)**.** In a maintenance department of a firm, the life of the electronic equipment is found to be normally distributed. A random sample of 30 sensors has a sample mean life of 800 hours and a sample standard deviation of 70 hours.

(i) Construct a 95% two-sided CI for the mean life of the sensors.

(ii) Construct a 99% two-sided CI for the variance of life of sensors.

(iii) If the population standard deviation of life of sensors is known to be 90 hours, construct a 99% two-sided CI for the mean life of the sensors.

(iv) Comment on the difference in answers to (i) and (iii).

Solution. Life $\sim N(\mu, \sigma^2)$, $\bar{x} = 800$ hr, $s = 70$ hr, $n = 30$.

(i) $\alpha = 0.05$, $t_{\alpha/2,n-1} = t_{0.025,29} = 2.045$:

$$\bar{x} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} = 800 \pm 2.045 \cdot \frac{70}{\sqrt{30}} = 800 \pm 26.135 \ \Rightarrow \ \boxed{[773.86,\ 826.14] \text{ hr.}}$$

(ii) 99% CI on the variance:

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}} = \frac{29 \cdot 70^2}{52.34} \text{ to } \frac{29 \cdot 70^2}{13.12} \ \Rightarrow \ \boxed{(2715,\ 10830.7).}$$

(iii) σ known $= 90$, 99% CI on the mean, $Z_{0.005} = 2.58$:

$$\bar{x} \pm Z_{0.005}\frac{\sigma}{\sqrt{n}} = 800 \pm 2.58 \cdot \frac{90}{\sqrt{30}} = 800 \pm 42.39 \ \Rightarrow \ \boxed{(757.6,\ 842.4) \text{ hr.}}$$

(iv) The confidence interval of (iii) is wider than that of (i), even though σ is known — because the confidence level in (iii) is 99% against 95% in (i).

Example 6.11 (Electronic sensor life — 99% CIs)**.** The life of an electronic sensor is normally distributed. A random sample of 20 sensors has a mean life of 1000 hours and a standard deviation of 100 hours.

(i) Construct a 99% two-sided CI for the mean life of the sensors.

(ii) Construct a 95% two-sided CI for the variance of life of sensors.

(iii) If the population standard deviation of the life of sensors is known to be 105 hours, construct a 99% two-sided CI for the mean life of the sensors.

(iv) Comment on the difference in answers to (i) and (iii).

Solution. Life $\sim N(\mu, \sigma^2)$, $\bar{x} = 1000$ hr, $s = 100$ hr, $n = 20$.

(i) 99% CL, two-sided, $\alpha = 0.01$, $t_{\alpha/2,\,n-1} = t_{0.005,19} = 2.861$:

$$\bar{x} \pm t_{\alpha/2,\,n-1}\frac{s}{\sqrt{n}} = 1000 \pm 2.861 \cdot \frac{100}{\sqrt{20}} \;\Rightarrow\; \boxed{[936.03,\ 1063.97] \text{ hr.}}$$

(ii) 95% CI on the variance, $\alpha = 0.05$, $\alpha/2 = 0.025$, $\chi^2_{0.025,19} = 32.85$, $\chi^2_{0.975,19} = 8.91$:

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}} = \frac{19 \cdot 100^2}{32.85} \text{ to } \frac{19 \cdot 100^2}{8.91}$$

$$\Rightarrow \boxed{(5783.87,\ 21324.35) \text{ (variance).}}$$

(iii) $\sigma = 105$ known, 99% CI on the mean, $\alpha = 0.01$, $Z_{0.005} = 2.58$:

$$\bar{x} \pm Z_{0.005}\frac{\sigma}{\sqrt{n}} = 1000 \pm 2.58 \cdot \frac{105}{\sqrt{20}} \;\Rightarrow\; \boxed{(939.42,\ 1060.58) \text{ hr.}}$$

(iv) The CI of (iii) is narrower than that of (i), since σ is known. A more precise interval is obtained when additional information is available.

Example 6.12 (Small sample — CI on the mean, variance unknown)**.** For the sample $(5, 2, 3, 1, 0, 0, 1, 3, 3)$ from a population, construct a 95% CI on the population mean if the population variance is unknown.

Solution. $\mu \in \bar{x} \pm t_{\alpha/2,\,n-1}\dfrac{s}{\sqrt{n}}$, with $\bar{X} = 2$, $n = 9$, $t_{0.025,8} = 2.306$, and

$$s^2 = \tfrac{1}{8}\big[(5-2)^2 + (2-2)^2 + (3-2)^2 + (1-2)^2 + (0-2)^2 + (0-2)^2 + (1-2)^2 + (3-2)^2 + (3-2)^2\big] = \frac{22}{8} = 2.75,$$

so $s = 1.658$. Then

$$2 \pm 2.306 \cdot \frac{1.658}{\sqrt{9}} \;\Rightarrow\; \boxed{[0.7253,\ 3.2747].}$$

Example 6.13 (Cloud seeding)**.** Cloud seeding has been studied for many decades as a weather modification procedure. The rainfall in acre-feet from 20 clouds that were selected at random and seeded with silver nitrate are as follows:

18.0, 30.7, 19.8, 27.1, 22.3, 18.8, 31.8, 23.4, 21.2, 27.9,
31.9, 27.1, 25.0, 24.7, 26.9, 21.8, 29.2, 34.8, 26.7, 31.6.

(a) Construct a 95% one-sided lower bound on the mean.

(b) Construct a 90% two-sided CI on the mean. What is the width of this interval?

(c) If the width of the interval in part (b) were to be halved, how much should be the sample size? Use trial and error if needed.

Solution. $n = 20$, $\bar{X} = 26.035$, $s = 4.7848$.

(a) 95% lower bound, $t_{0.05,19} = 1.729$:

$$\bar{X} - t_{0.05,19}\frac{s}{\sqrt{n}} = 26.035 - 1.729 \cdot \frac{4.7848}{\sqrt{20}} \Rightarrow \boxed{\mu \geq 24.185.}$$

(b) 90% two-sided CI, $\alpha = 0.10$, $\alpha/2 = 0.05$, $t_{0.05,19} = 1.729$:

$$\bar{X} \pm t_{\alpha/2,\,n-1}\frac{s}{\sqrt{n}} = 26.035 \pm 1.729 \cdot \frac{4.7848}{\sqrt{20}} \Rightarrow \boxed{(24.185,\ 27.885).}$$

Width $=$ max $-$ min $= 3.7958$.

(c) We require width $= 2\,t_{0.05,\,n-1}\dfrac{s}{\sqrt{n}}$ to be halved, i.e. $3.7958/2 = 1.8979$ (assuming s remains). By trial and error:

$$n = 30: \ \text{width} = 2 \cdot 1.697 \cdot \frac{4.7848}{\sqrt{30}} = 2.96;$$

$$n = 40: \ \text{width} = 2 \cdot 1.684 \cdot \frac{4.7848}{\sqrt{40}} = 2.5;$$

$$n = 60: \ \text{width} = 2 \cdot 1.671 \cdot \frac{4.7848}{\sqrt{60}} = 2.06;$$

$$n = 120: \ \text{width} = 2 \cdot 1.658 \cdot \frac{4.7848}{\sqrt{120}} = 1.4484.$$

Since the target width 1.8979 lies between the widths at $n = 60$ and $n = 120$, we conclude $\boxed{60 \leq n \leq 120.}$

Example 6.14 (Gasoline prices and highway speeds)**.** (a) From a sample of 42 gasoline stations statewide, the average price of a gallon of unleaded gas was found to be \$1.12 and the standard deviation was \$0.04 per gallon. Construct a 99% CI on the mean price of a gallon of unleaded gas.

(b) A random sample of 57 vehicles on a highway is clocked with an average speed of 23.2 mph and a standard deviation of 0.3 mph. Estimate (i) the standard deviation of the population and (ii) the standard error of the mean for this population.

Solution.

(a) $n = 42$, $\bar{x} = 1.12$, $s = 0.04$, $\alpha = 0.01$, $t_{0.01/2,\,41} = t_{0.005,41} = 2.7$:

$$\bar{x} \pm t_{\alpha/2,\,n-1}\frac{s}{\sqrt{n}} = 1.12 \pm 2.7 \cdot \frac{0.04}{\sqrt{42}} \Rightarrow \boxed{99\%\ \text{CI} = [\$1.10,\ \$1.14].}$$

(b) $n = 57$, $\bar{x} = 23.2$, $s = 0.3$, assume $\alpha = 0.05$. *(i)* A 95% CI for σ^2:

$$\left(\frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}},\ \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}}\right) = \left(\frac{56 \cdot 0.3^2}{77.4},\ \frac{56 \cdot 0.3^2}{36.5}\right) = (0.065,\ 0.138),$$

so the 95% CI for σ is $(0.255,\ 0.37)$. *(ii)* The standard error of the mean is estimated as

$$\frac{(0.255 + 0.37)/2}{\sqrt{57}} = 0.041.$$

Example 6.15 (Contamination level in food packs)**.** The level of contamination in % was studied in 6 packs of food with the following data:

$$16.8,\ 17.2,\ 17.4,\ 16.9,\ 16.5,\ 17.1.$$

Find a 99% confidence interval on the mean % μ, and provide a practical interpretation of this interval.

Solution.

$$\bar{X} = \frac{101.9}{6} = 16.98,$$

$$s^2 = \frac{0.0324 + 0.0484 + 0.1764 + 0.006 + 0.2304 + 0.0144}{5} = 0.1016, \qquad s = 0.318.$$

99% CI, $t_{0.01/2,\,5} = t_{0.005,5} = 4.032$:

$$\bar{x} \pm t_{0.005,5}\frac{s}{\sqrt{n}} = 16.98 \pm 4.032 \cdot \frac{0.318}{\sqrt{6}} = 16.98 \pm 0.523 \ \Rightarrow\ \boxed{[16.457,\ 17.503].}$$

Example 6.16 (Glass-bottle wall thickness — lower bound)**.** The wall thickness of 25 glass bottles was measured, and the sample mean was 4.05 mm and the sample standard deviation was 0.08 mm. Find a 95% lower confidence bound for the mean wall thickness, and interpret the interval.

Solution. $n = 25$, $\bar{x} = 4.05$, $s = 0.08$, $\alpha = 0.05$, $t_{0.05,24} = 1.711$:

$$\mu \geq \bar{X} - t_{\alpha,\,n-1}\frac{s}{\sqrt{n}} = 4.05 - 1.711 \cdot \frac{0.08}{\sqrt{25}} \ \Rightarrow\ \boxed{\mu \geq 4.02\,\text{mm}.}$$

The minimum population mean wall thickness is 4.02 mm.

6.6 Confidence Interval on the Variance and Standard Deviation

The χ^2 distribution

Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal distribution with mean μ and variance σ^2, and let s^2 be the sample variance. Then the random variable

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

has a chi-square (χ^2) distribution with $n-1$ degrees of freedom.

CI on the Variance

If s^2 is the sample variance from a random sample of n observations from a normal distribution with unknown variance σ^2, then a $100(1-\alpha)\%$ confidence interval on σ^2 is

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}},$$

where $\chi^2_{\alpha/2,\,n-1}$ and $\chi^2_{1-\alpha/2,\,n-1}$ are the upper and lower $100\alpha/2\%$ points of the chi-square distribution with $n-1$ degrees of freedom respectively. A confidence interval for σ has lower and upper limits that are the square roots of the corresponding limits for σ^2.

One-Sided Confidence Bounds on the Variance

The $100(1-\alpha)\%$ lower and upper confidence bounds on σ^2 are

$$\frac{(n-1)s^2}{\chi^2_{\alpha,\,n-1}} \le \sigma^2 \qquad \text{and} \qquad \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha,\,n-1}}.$$

6.7 Worked Problems: Variance and Standard Deviation

Example 6.17 (Reading χ^2 percentiles)**.** Determine the values of the following percentiles: $\chi^2_{0.05,10}$, $\chi^2_{0.95,16}$, $\chi^2_{0.005,25}$.

Solution.

$$\chi^2_{0.05,10} = 18.31, \qquad \chi^2_{0.95,16} = 5.14, \qquad \chi^2_{0.005,25} = 46.93.$$

Example 6.18 (χ^2 percentiles for CIs)**.** Determine the χ^2 percentile that is required to construct each of the following CIs.

(a) Confidence level $= 95\%$, dof $= 24$, one-sided (upper).

(b) Confidence level $= 99\%$, dof $= 9$, one-sided (lower).

(c) Confidence level $= 90\%$, dof $= 19$, two-sided.

Solution.

(a) $\chi^2_{0.05,24} = 36.42$.

(b) $\chi^2_{1-0.01,9} = \chi^2_{0.99,9} = 2.09$.

(c) $\chi^2_{0.1/2,\,19} = \chi^2_{0.05,19} = 30.14$.

Example 6.19 (Rivet hole diameter — lower bound on variance)**.** A rivet is to be inserted into a hole. A random sample of $n = 15$ parts is selected, and the hole diameter is measured. The sample standard deviation of the hole diameter measurement is $s = 0.008$ mm. Construct a 99% lower confidence bound for the variance.

Solution. $n = 15$, $s = 0.008$. Lower confidence bound $\dfrac{(n-1)s^2}{\chi^2_{\alpha,\,n-1}} \le \sigma^2$, with $\chi^2_{0.01,14} = 29.14$:

$$\sigma^2 \ge \frac{14 \cdot 0.008^2}{29.14} = 3.074 \times 10^{-5} \;\Rightarrow\; \boxed{\text{CI} = [0.00003074,\ \infty).}$$

Example 6.20 (Sugar content of canned peaches)**.** The sugar content of the syrup in canned peaches is normally distributed. A random sample of $n = 10$ cans yields a sample standard deviation of $s = 4.8$ mg. Calculate a 95% two-sided CI for σ.

Solution. $n = 10$, $s = 4.8$ mg, CL $= 95\%$, $\alpha = 0.05$, $\chi^2_{0.025,9} = 19.02$, $\chi^2_{0.975,9} = 2.70$:

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}} = \frac{9 \cdot 4.8^2}{19.02} \text{ to } \frac{9 \cdot 4.8^2}{2.70}$$

$$\Rightarrow\ 10.90 \le \sigma^2 \le 76.8 \ \Rightarrow\ \sqrt{10.9} \le \sigma \le \sqrt{76.8} \ \Rightarrow\ \boxed{3.30 \le \sigma \le 8.76.}$$

Example 6.21 (CI on population variance — small sample)**.** For the sample $(5, 2, 3, 1, 0, 0, 1, 3, 3)$ from a population, construct a 95% confidence interval on the population variance.

Solution. $s = 1.6583$, $n = 9$, $\chi^2_{0.05,8} = 15.51$, $\chi^2_{0.95,8} = 2.73$:

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}} = \frac{8 \cdot 1.6583^2}{15.51} \text{ to } \frac{8 \cdot 1.6583^2}{2.73}$$

$$\Rightarrow\ 1.4189 \le \sigma^2 \le 8.058\ \Rightarrow\ \boxed{1.19 \le \sigma \le 2.838.}$$

Example 6.22 (Amino acid concentration in dog feeds)**.** Amino acid concentrations in 8 packs of dog feed were recorded in mg/kg of dog feed as follows:

$$22,\ 24,\ 20,\ 26,\ 27,\ 24,\ 26,\ 23.$$

(i) Perform a test at 5% significance to see if the population variance is greater than 5.5 $(\text{mg/kg})^2$. What is the P-value of the test?

(ii) Construct an upper and a lower 90% confidence bound for the population variance.

Solution. $H_0 : \sigma^2 = 5.5$, $H_1 : \sigma^2 > 5.5$ $(\text{mg/kg})^2$, $\alpha = 0.05$, $n = 8$, $\bar{X} = 24$, $s = 2.3299$.

$$s^2 = \frac{4+0+16+4+9+0+4+1}{7} = \frac{38}{7} = 5.428, \qquad s = \sqrt{5.428} = 2.3299.$$

Test statistic:

$$\chi^2_0 = \frac{(n-1)s^2}{\sigma^2_0} = \frac{7 \cdot 2.3299^2}{5.5} = 6.9091,$$

and $\chi^2_{\text{tab}} = \chi^2_{0.05,7} = 14.07$. Since $\chi^2_0 < \chi^2_{\text{tab}}$, do not reject H_0.

P-value: $P = \text{P}(\chi^2_7 > 6.91)$, which is between 0.4 and 0.5 (since $\text{P}(\chi^2_7 > 6.346) = 0.5$ and $\text{P}(\chi^2_7 > 7.283) = 0.4$).

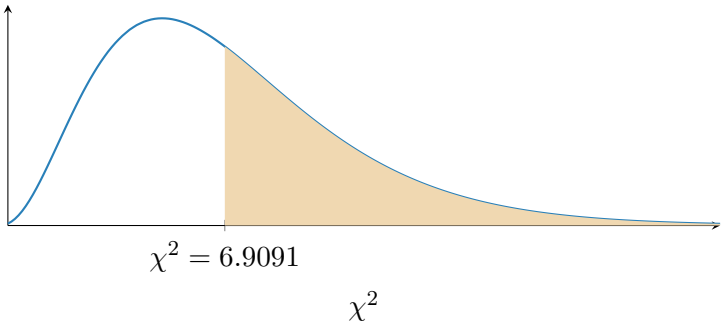


Figure 6.1: Chi-square distribution with 7 degrees of freedom; the shaded upper tail beyond $\chi^2 = 6.9091$ gives the P-value $(0.4 < P < 0.5)$.

(ii) 90% confidence bounds on the variance. Upper bound:

$$\sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha,\,n-1}} = \frac{7 \cdot 5.4286}{2.833} = 13.41.$$

Lower bound:

$$\sigma^2 \ge \frac{(n-1)s^2}{\chi^2_{\alpha,\,n-1}} = \frac{7 \cdot 5.4286}{12.02} = 3.16.$$

$$\therefore\ \boxed{3.16 \le \sigma^2 \le 13.41.}$$

Example 6.23 (Population standard deviation — two-sided and one-sided)**.** (Continuation.) For the same sample:

(a) Find a 90% two-sided confidence interval for the population standard deviation.

(b) Find a 95% upper bound and a 95% lower bound for the population standard deviation. How do these bounds compare with the limits of part (a)?

Solution. $n = 10$, $s = 4.8$ (data of the peaches problem).

(a) $\alpha = 0.10$, two-sided, $\alpha/2 = 0.05$, $\chi^2_{0.05,9} = 16.92$, $\chi^2_{0.95,9} = 3.325$:

$$90\%\text{ CI for } \sigma^2: \left(\frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}},\ \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}}\right) = \left(\frac{9\cdot 4.8^2}{16.92},\ \frac{9\cdot 4.8^2}{3.325}\right) = (12.26,\ 62.36).$$

Interval for σ: $\boxed{(3.5,\ 7.9).}$

(b) 95% upper bound, $\alpha = 0.05$, $1-\alpha = 0.95$, $\chi^2_{0.95,9} = 3.325$:

$$\sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha,\,n-1}} = \frac{9\cdot 4.8^2}{3.325} = 62.36 \ \Rightarrow\ \sigma \le 7.9.$$

95% lower bound, $\chi^2_{0.05,9} = 16.92$:

$$\sigma^2 \ge \frac{(n-1)s^2}{\chi^2_{\alpha,\,n-1}} = \frac{9\cdot 4.8^2}{16.92} = 12.26 \ \Rightarrow\ \sigma \ge 3.5.$$

Both bounds equal the limits of the two-sided CI, since $\alpha/2$ in part (a) equals α in part (b).

Example 6.24 (Hole diameter — 99% lower bound on variance)**.** A hole diameter is measured in a random sample of 15 parts. The sample standard deviation of the hole diameter is 0.008 mm. Construct a 99% lower confidence bound for σ^2.

Solution. $n = 15$, $s = 0.008$, 99% CI $\Rightarrow \alpha = 0.01$, $\chi^2_{0.01,14} = 29.14$:

$$\sigma^2 \ge \frac{(n-1)s^2}{\chi^2_{\alpha,\,n-1}} = \frac{14\cdot 0.008^2}{29.14} = 3.84\times 10^{-3} \ \Rightarrow\ \boxed{\sigma \ge 0.062.}$$

Example 6.25 (Weight measurements — test and lower bound on σ)**.** A set of 15 weight measurements gives sample standard deviation $s = 0.0088$ gm.

(i) Does the population standard deviation σ of all the weight measurements exceed 0.008 gm at 5% significance level? Find the P-value for this test.

(ii) Answer part (i) with the help of a lower confidence interval for σ with $\alpha = 5\%$.

Solution. $n = 15$, $s = 0.0088$ gm, $H_0 : \sigma = 0.008$ gm, $H_1 : \sigma > 0.008$ gm, $\alpha = 0.05$.

(i)

$$\chi^2_0 = \frac{(n-1)s^2}{\sigma^2} = \frac{14\cdot 0.0088^2}{0.008^2} = 16.94.$$

From tables, $\chi^2_{\text{table}} = \chi^2_{0.05,14} = 23.68$. Since $\chi^2_0 < \chi^2_{\text{table}}$, we fail to reject H_0, i.e. the standard deviation does not exceed 0.008 gm at 5% significance. *P-value:* $\chi^2_0 = 16.94$ lies between 0.5 and 0.1 (i.e. $21.06 < \chi^2$-range< 13.34), so $0.1 < P < 0.5$.

(ii) Lower confidence bound, $\alpha = 0.05$, $\chi^2_{0.05,14} = 23.68$:

$$\sigma^2 \geq \frac{(n-1)s^2}{\chi^2_{\alpha,\,n-1}} = \frac{14 \cdot 0.0088^2}{23.68} = 4.578 \times 10^{-5} \;\Rightarrow\; \boxed{\sigma \geq 6.76 \times 10^{-3} \approx 0.0067.}$$

Since the lower bound $0.0067 < 0.008$, the interval is consistent with failing to reject H_0.

Example 6.26 (Titanium percentage in an aerospace alloy)**.** The % of titanium in an alloy used in aerospace casting is measured in 51 randomly selected parts. The sample standard deviation is $s = 0.28$. Construct a 95% two-sided confidence interval for σ.

Solution. $n = 51$, $s = 0.28$, $\alpha = 0.05$, $\chi^2_{0.025,50} = 71.42$, $\chi^2_{0.975,50} = 32.36$:

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}} = \frac{50 \cdot 0.28^2}{71.42} \text{ to } \frac{50 \cdot 0.28^2}{32.36}$$

$$\Rightarrow\; 0.055 \leq \sigma^2 \leq 0.121.$$

Taking the square root of the endpoints of this interval,

$$\boxed{0.234 \leq \sigma \leq 0.348.}$$

Compiler's Note on Fidelity and Coverage

This textbook was reconstructed directly from the scanned handwritten notes (76 pages). Wherever the handwriting was legible, the content — every definition, formula, worked problem and final numerical answer — was reproduced one-to-one. The original's final answers are treated as authoritative; only intermediate working that was ambiguous in the scan was reconstructed so as to reach those answers. No topics were invented and no problems were fabricated.

Coverage

All six chapters cover the full manuscript: hypothesis-testing fundamentals (Ch. 1); tests on the mean with variance known (Ch. 2); tests on the variance (Ch. 3); tests on the mean with variance unknown, the t-test (Ch. 4); goodness-of-fit tests, including the Poisson, binomial and uniform (fair-die) examples (Ch. 5); and the analysis of statistical intervals — confidence intervals on the mean (variance known and unknown) and on the variance/standard deviation, with sample-size selection and one-sided bounds (Ch. 6).

Points reconstructed, clarified, or carrying an apparent slip in the source

- **All normal / t / χ^2 curves** are schematic TikZ/pgfplots redrawings of the hand-sketched bell curves and rejection regions; shaded areas mark the critical or β regions described in the text. Exact sketch widths are illustrative, but every labelled critical value, tail area, and axis mark matches the notes.
- **Ch. 2, "P-values for a two-sided test", case (b)** $Z_0 = -1.84$. The manuscript records 0.64668, but $2[1 - \Phi(1.84)] = 2(0.032884) = 0.0658$; the notes' value appears to be a decimal slip. Both are shown.
- **Ch. 2, battery-life sample size** evaluates to $n \approx 0.844$, rounded up to $n \approx 1$ in the notes; reproduced exactly as written (physically trivial).
- **Ch. 3, ice-hockey σ problem** momentarily mixes $n = 16$ and $n = 17$ while forming the bound; the tabulated values used ($\chi^2_{0.05,16} = 26.30$) and the boxed final answers ($0.004 \le \sigma^2$, $0.07 \le \sigma$) are kept as in the source.
- **Ch. 5, goodness-of-fit tables** reproduce the notes' pooling of low-expected-frequency classes and their χ^2_{cal} values (2.94, 2.2935, 2.6859, 4.0, 5.8545) and degrees-of-freedom bookkeeping

exactly, including the radar-component example's use of $\chi^2_{0.05,4-1-1} = \chi^2_{0.05,2} = 5.99$.

- **Ch. 6, several CI problems** carry rounding in the source (e.g. $s = 1.449$ vs. $s = 1.658$ for the same 9-point sample in different problems, and the "overfill" variance CI). Each problem's own stated intermediate values and boxed answers are reproduced as written; where the source pools or rounds, the book follows the source rather than silently "correcting" it.
- **Ch. 6, standard-error-of-the-mean estimate** in the gasoline/speed problem follows the notes' own (non-standard) averaging of the σ CI endpoints to obtain 0.041.